

BFS Traversal

Step 1: Start at E

Legend

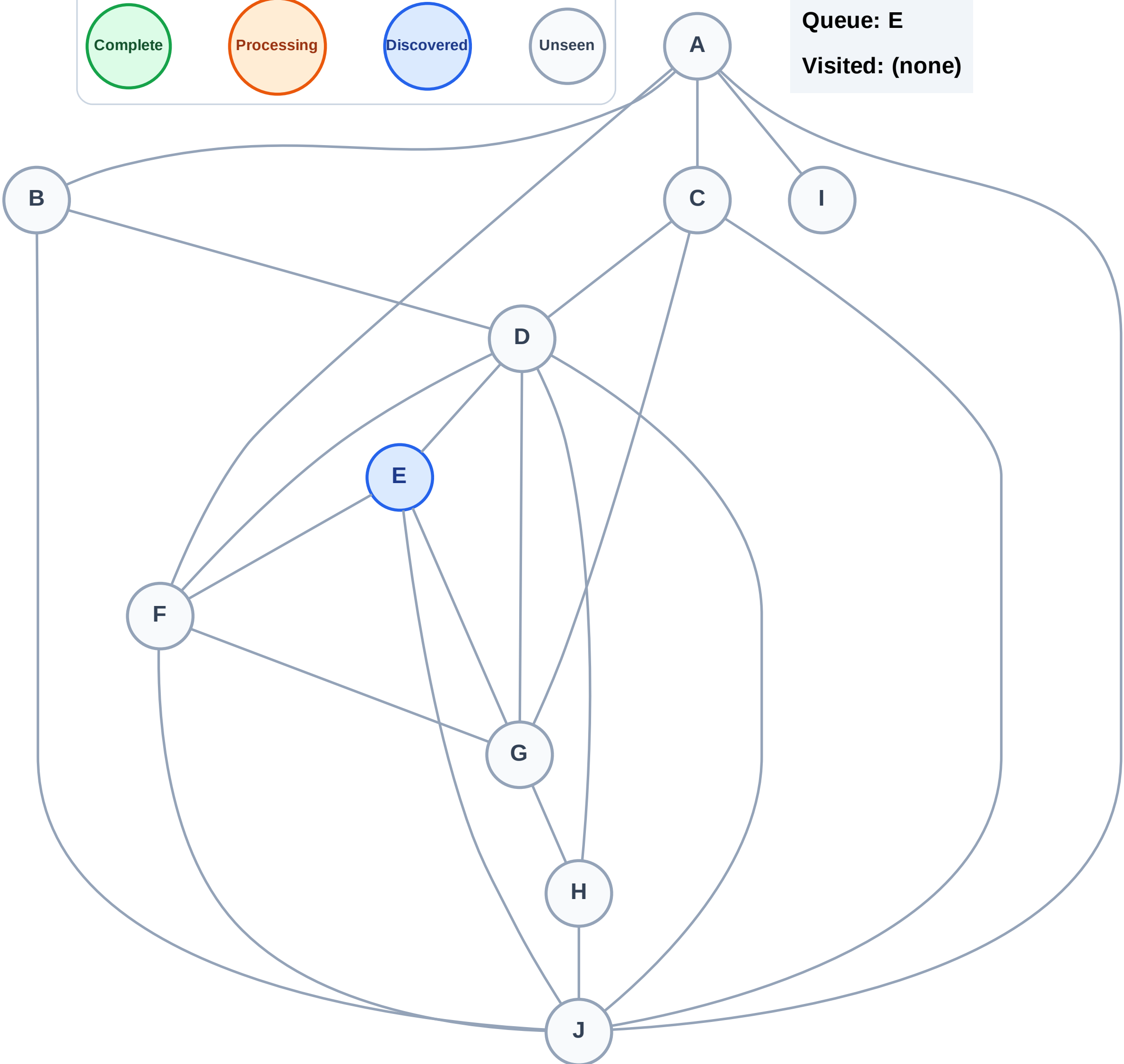
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

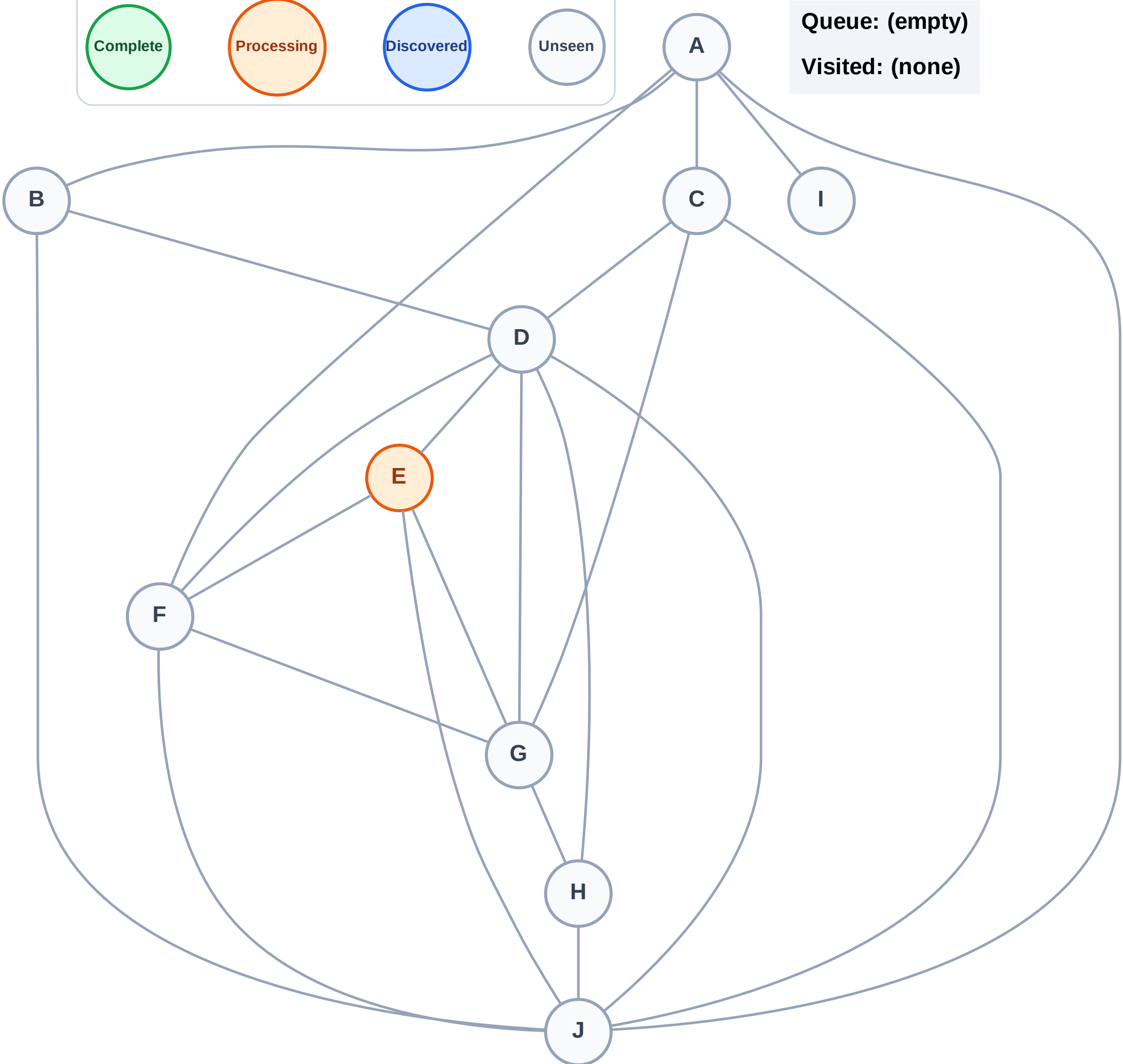
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

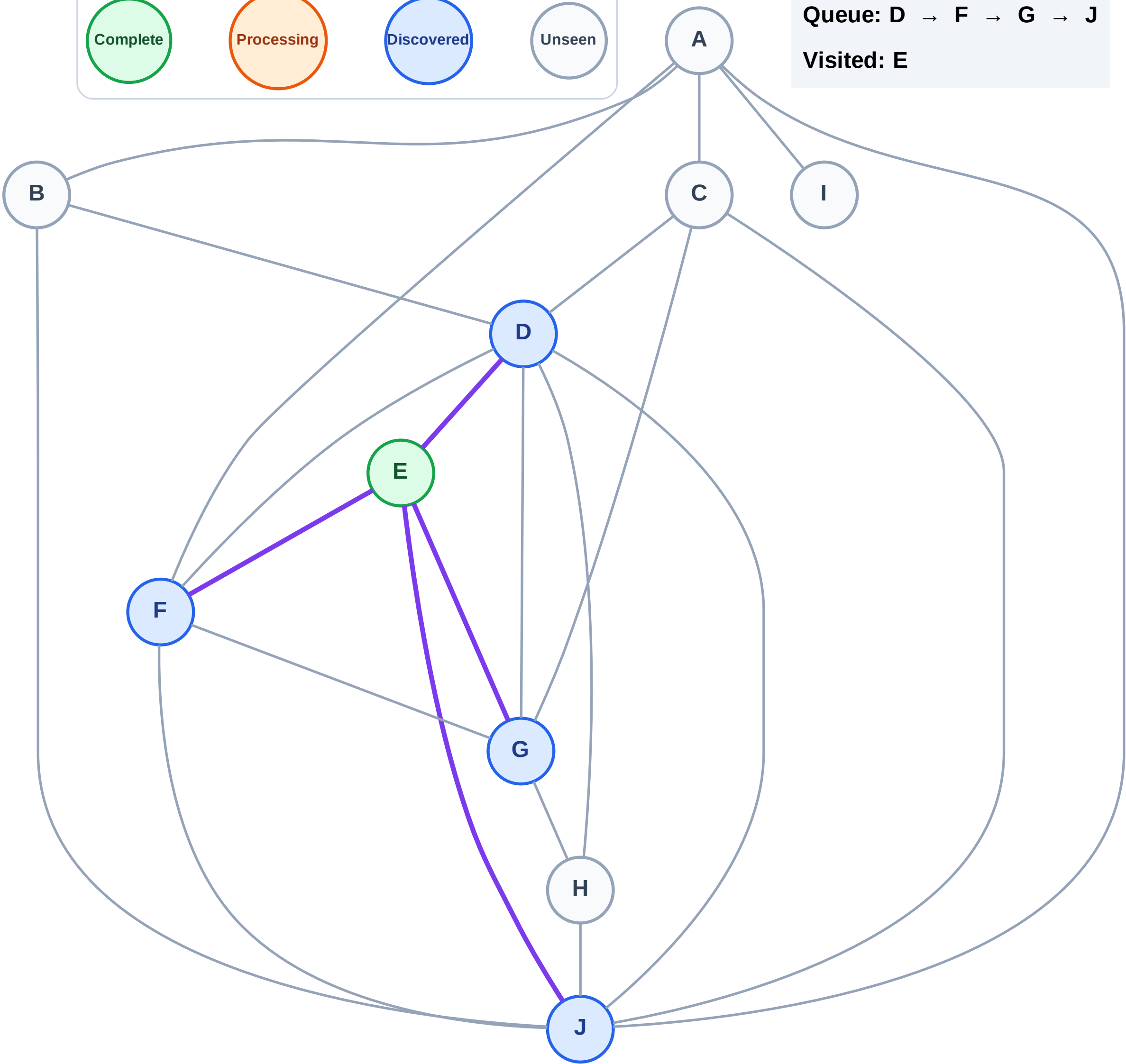
Step 3: Discover D, F, G, J from E

Legend



Queue: D → F → G → J

Visited: E



BFS Traversal

Step 4: Process node D

Legend

Complete

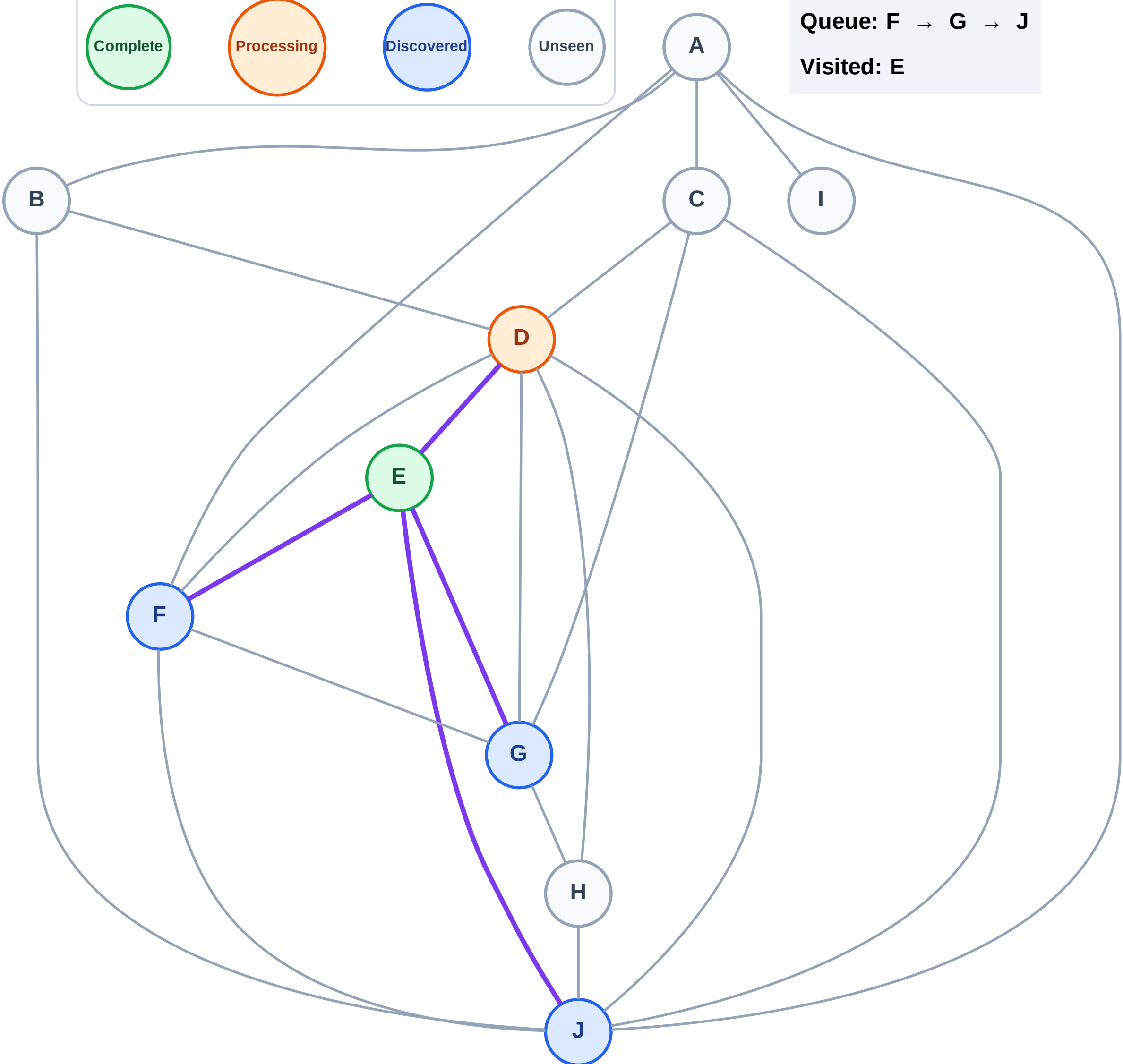
Processing

Discovered

Unseen

Queue: F → G → J

Visited: E



BFS Traversal

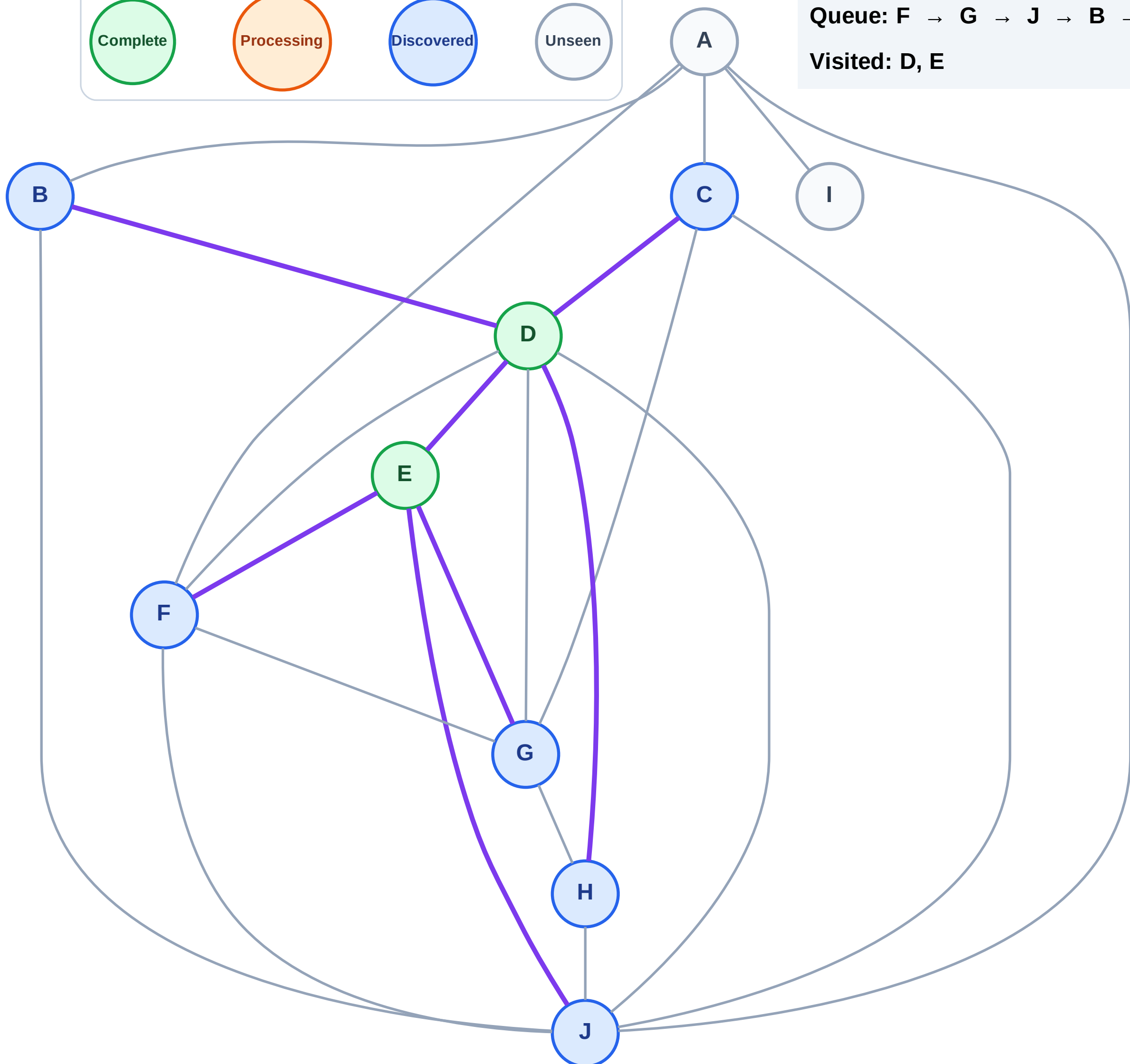
Step 5: Discover B, C, H from D

Legend



Queue: F → G → J → B → C → H

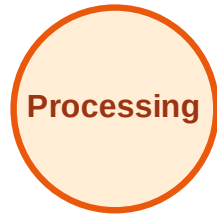
Visited: D, E



BFS Traversal

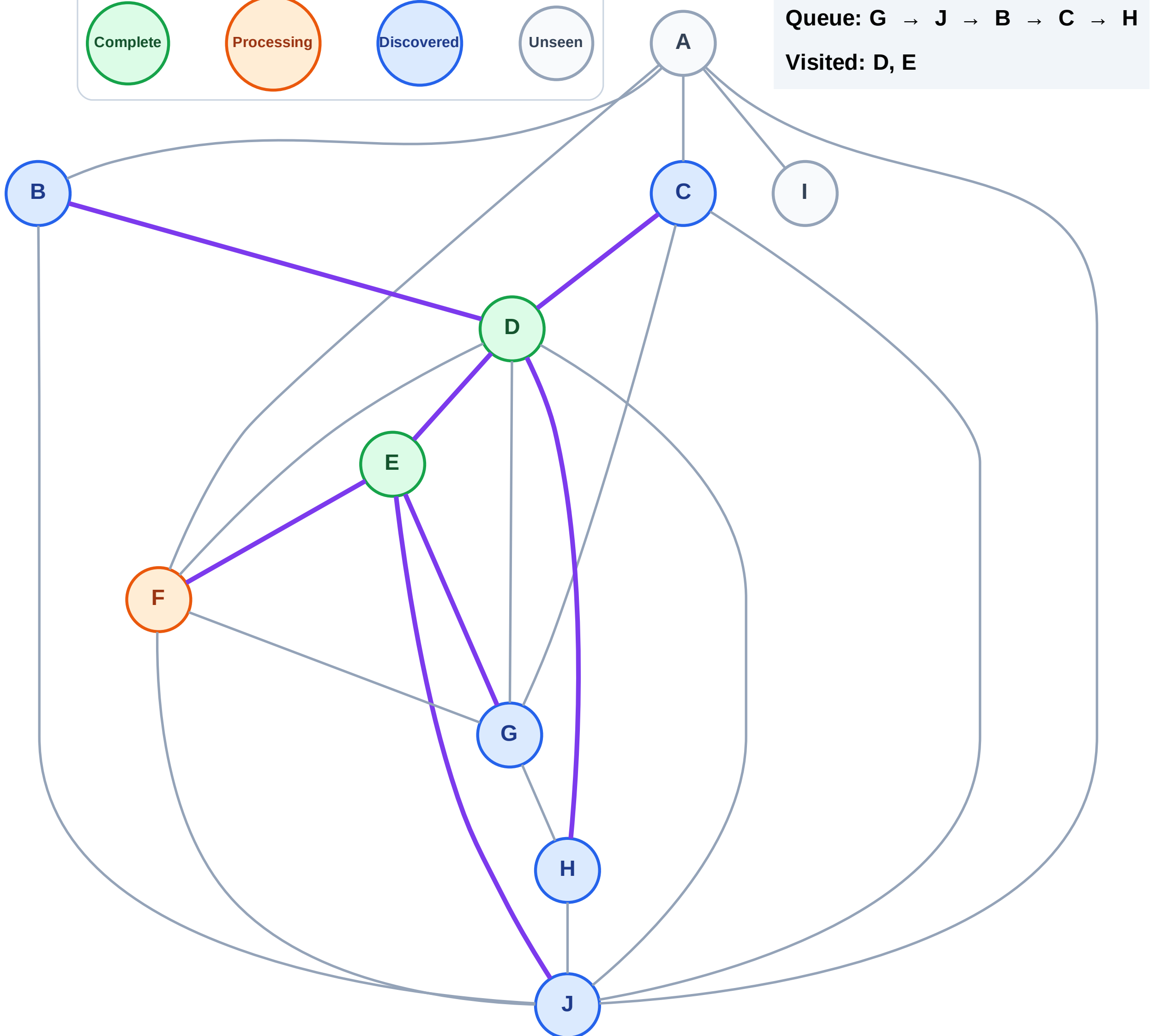
Step 6: Process node F

Legend



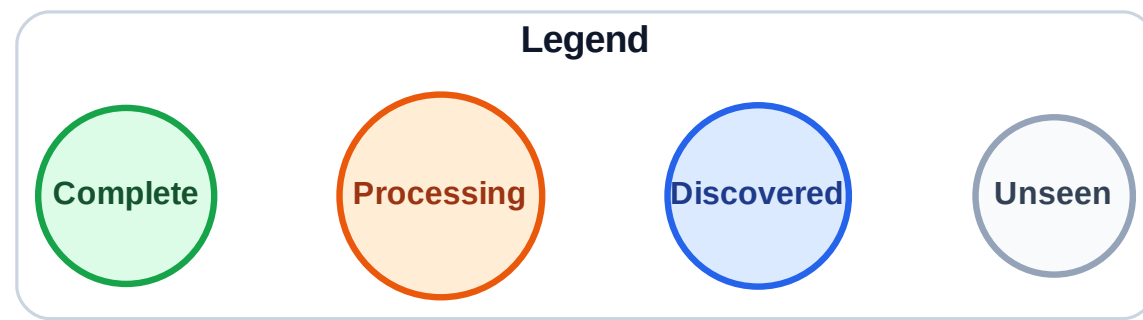
Queue: G → J → B → C → H

Visited: D, E

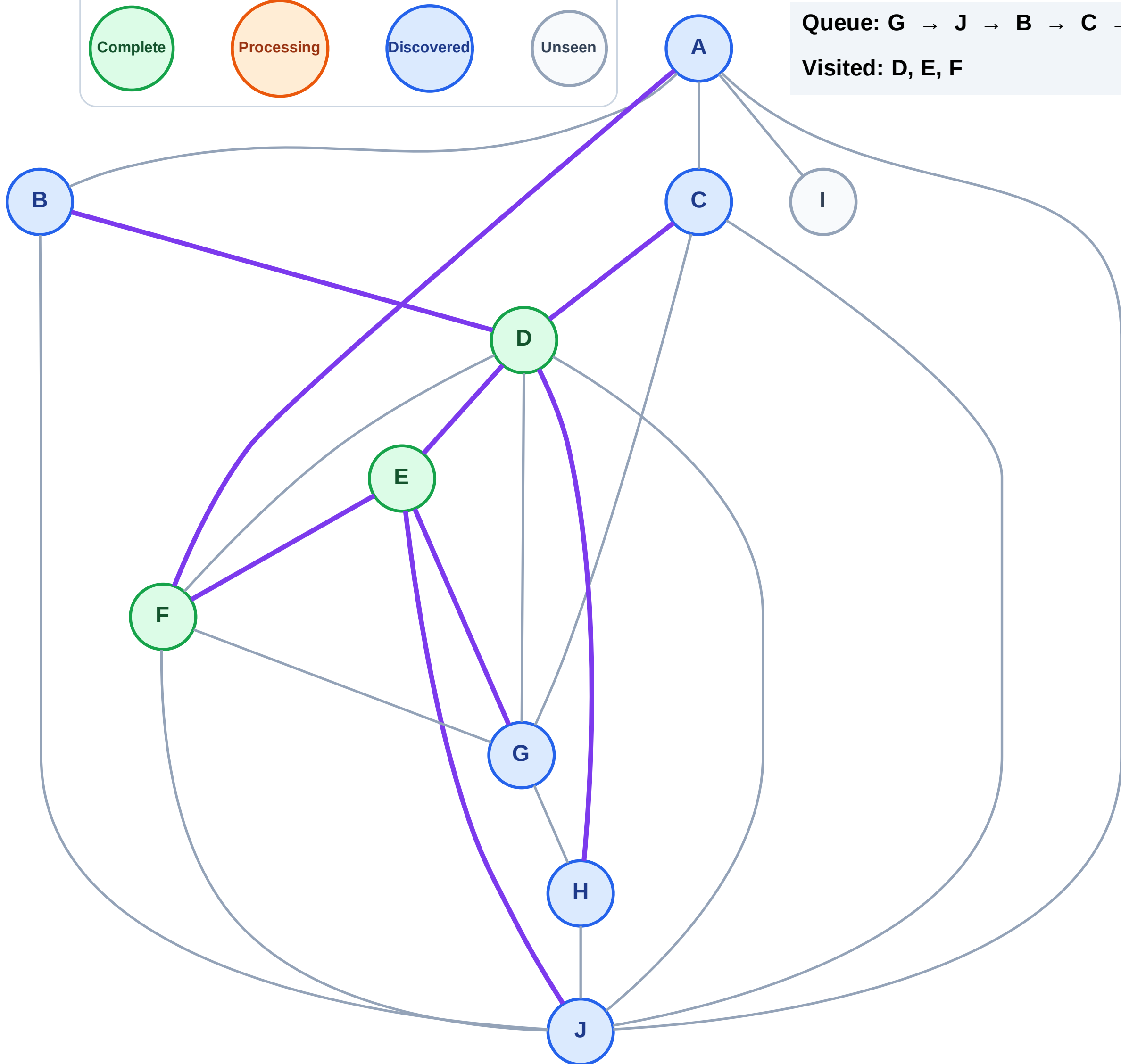


BFS Traversal

Step 7: Discover A from F



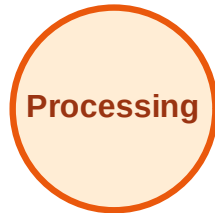
Queue: G → J → B → C → H → A
Visited: D, E, F



BFS Traversal

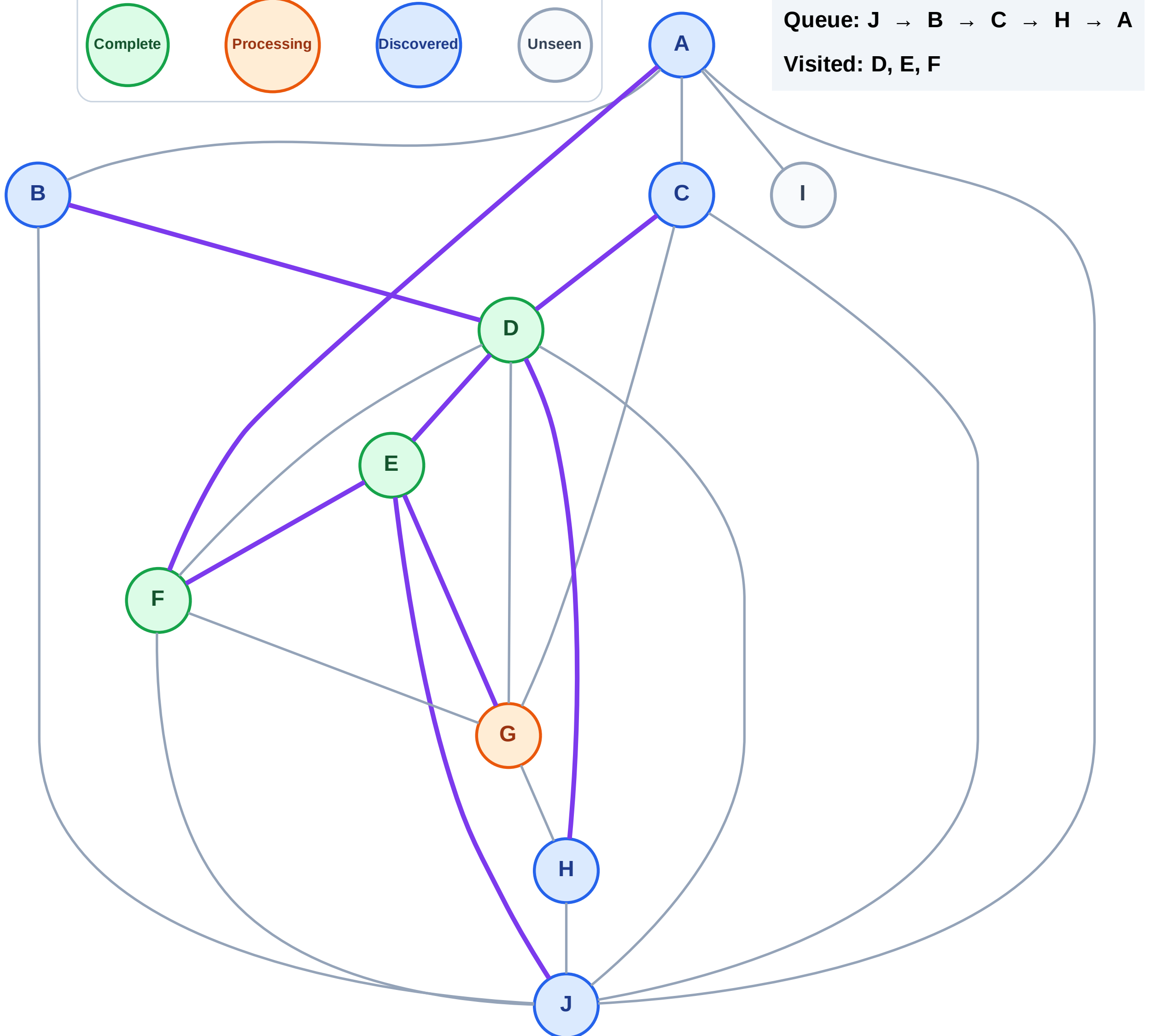
Step 8: Process node G

Legend



Queue: J → B → C → H → A

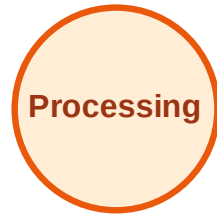
Visited: D, E, F



BFS Traversal

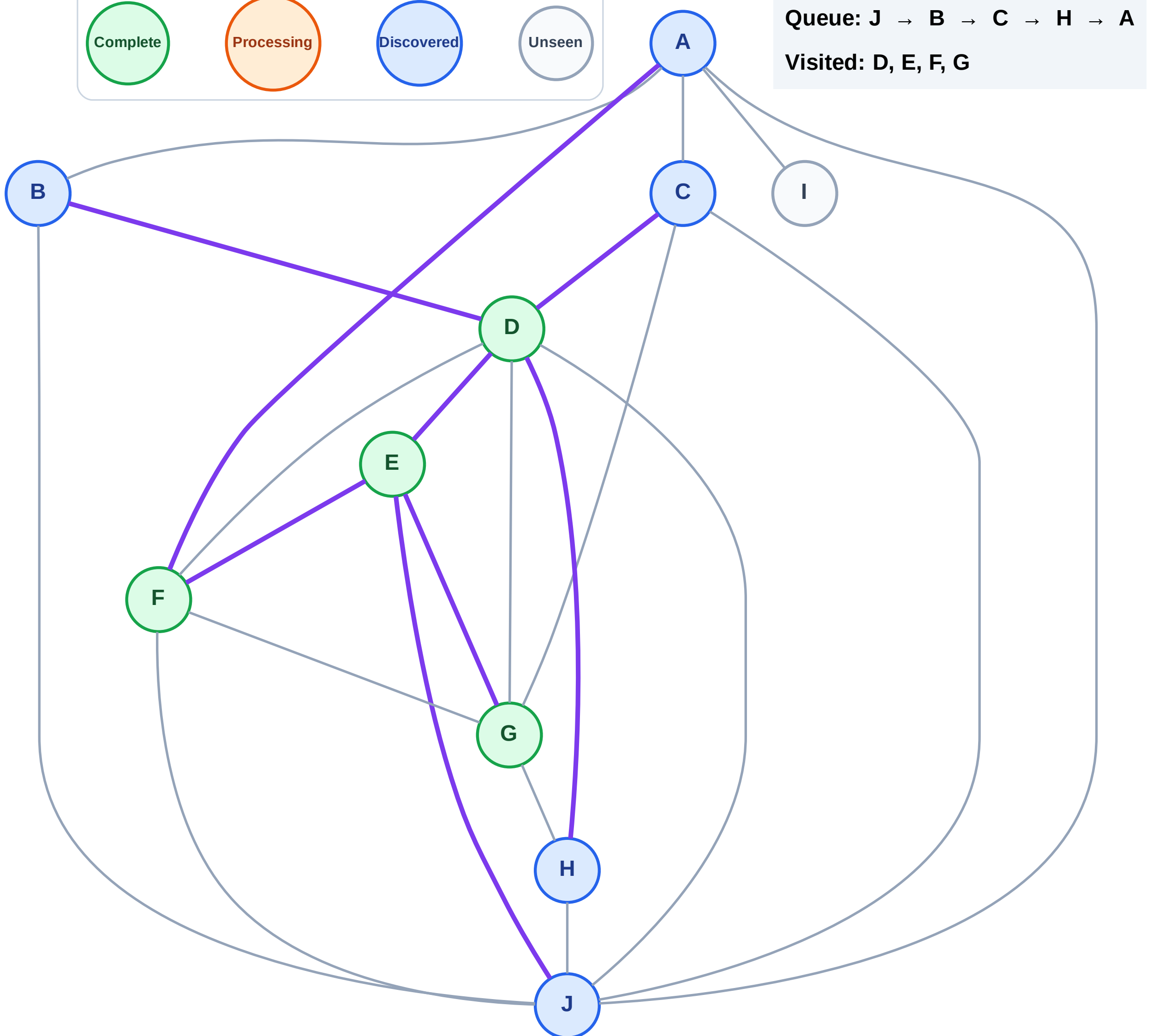
Step 9: No new neighbors from G

Legend



Queue: J → B → C → H → A

Visited: D, E, F, G



BFS Traversal

Step 10: Process node J

Legend

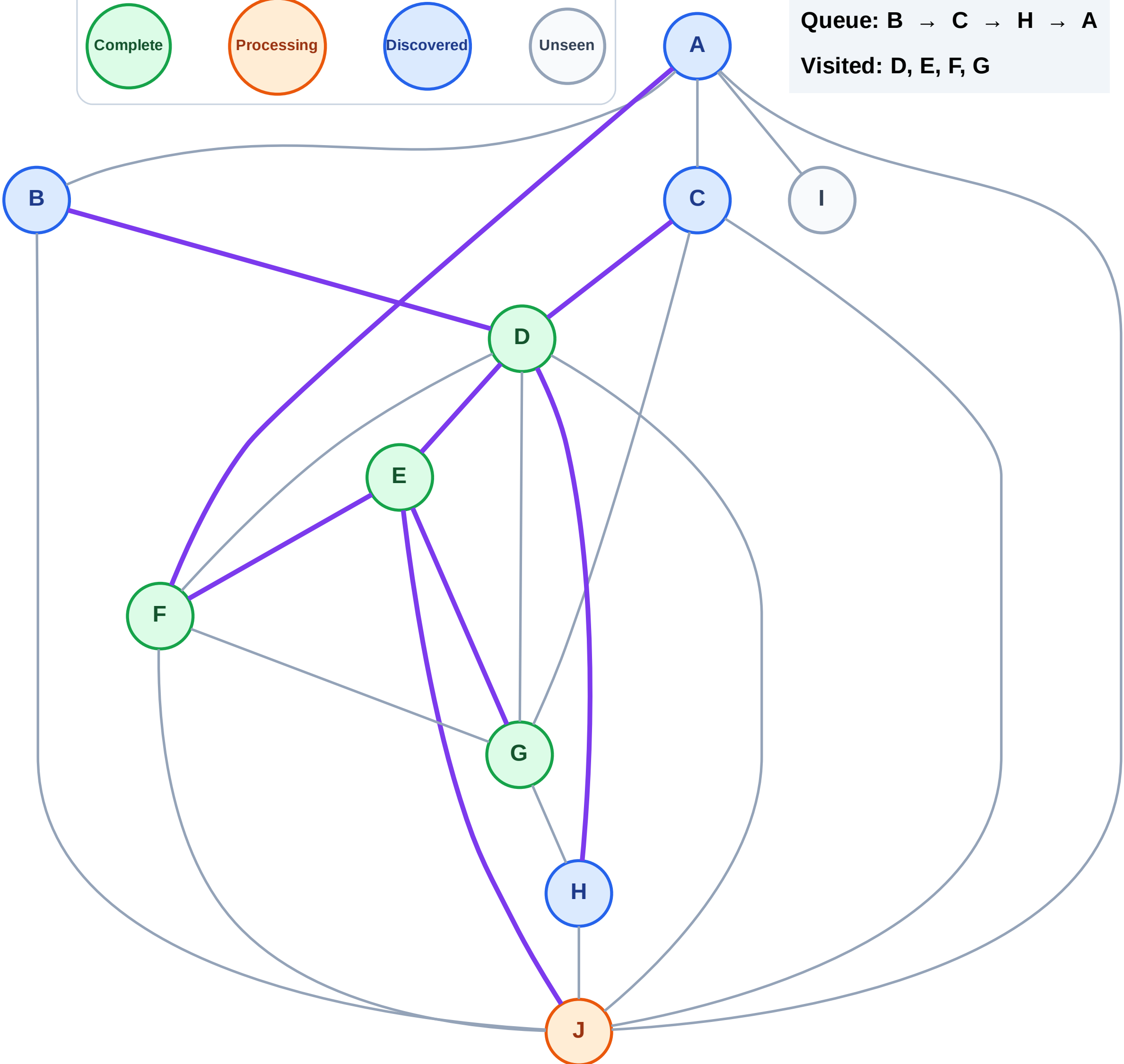
Complete

Processing

Discovered

Unseen

Queue: B → C → H → A
Visited: D, E, F, G



BFS Traversal

Step 11: No new neighbors from J

Legend

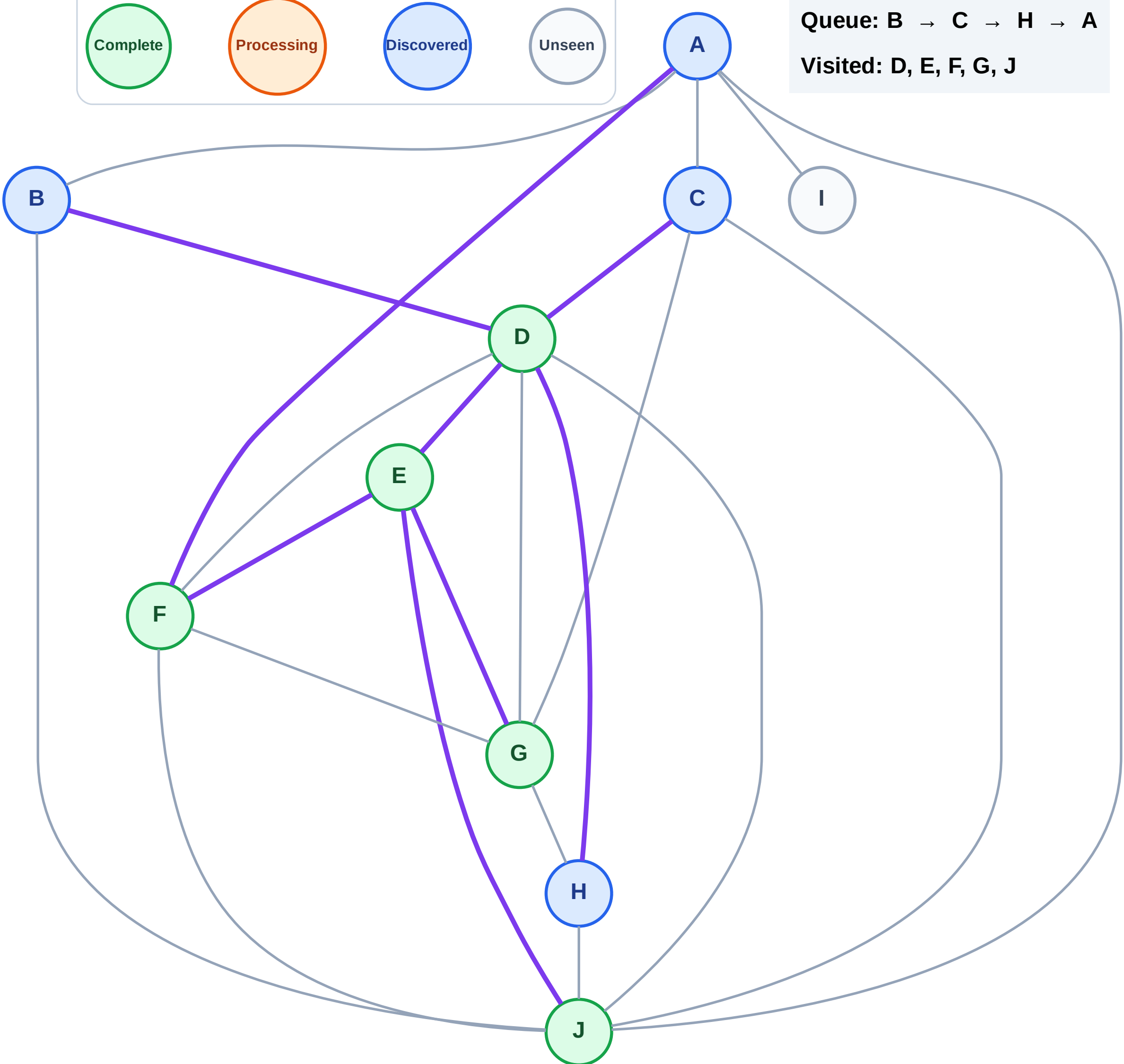
Complete

Processing

Discovered

Unseen

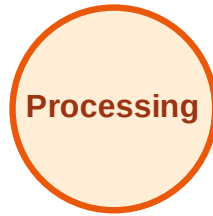
Queue: B → C → H → A
Visited: D, E, F, G, J



BFS Traversal

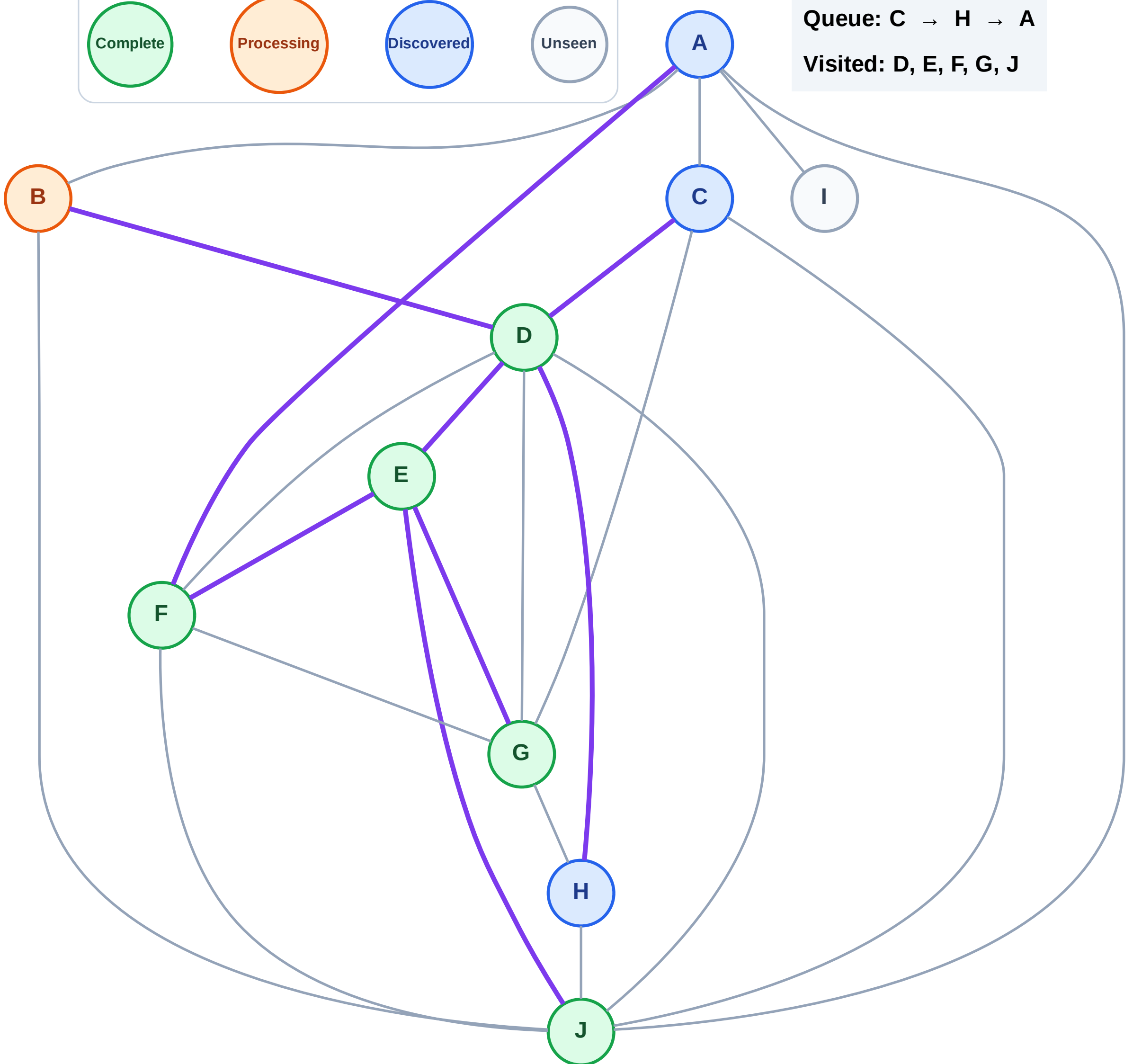
Step 12: Process node B

Legend



Queue: C → H → A

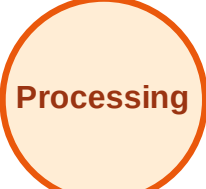
Visited: D, E, F, G, J



BFS Traversal

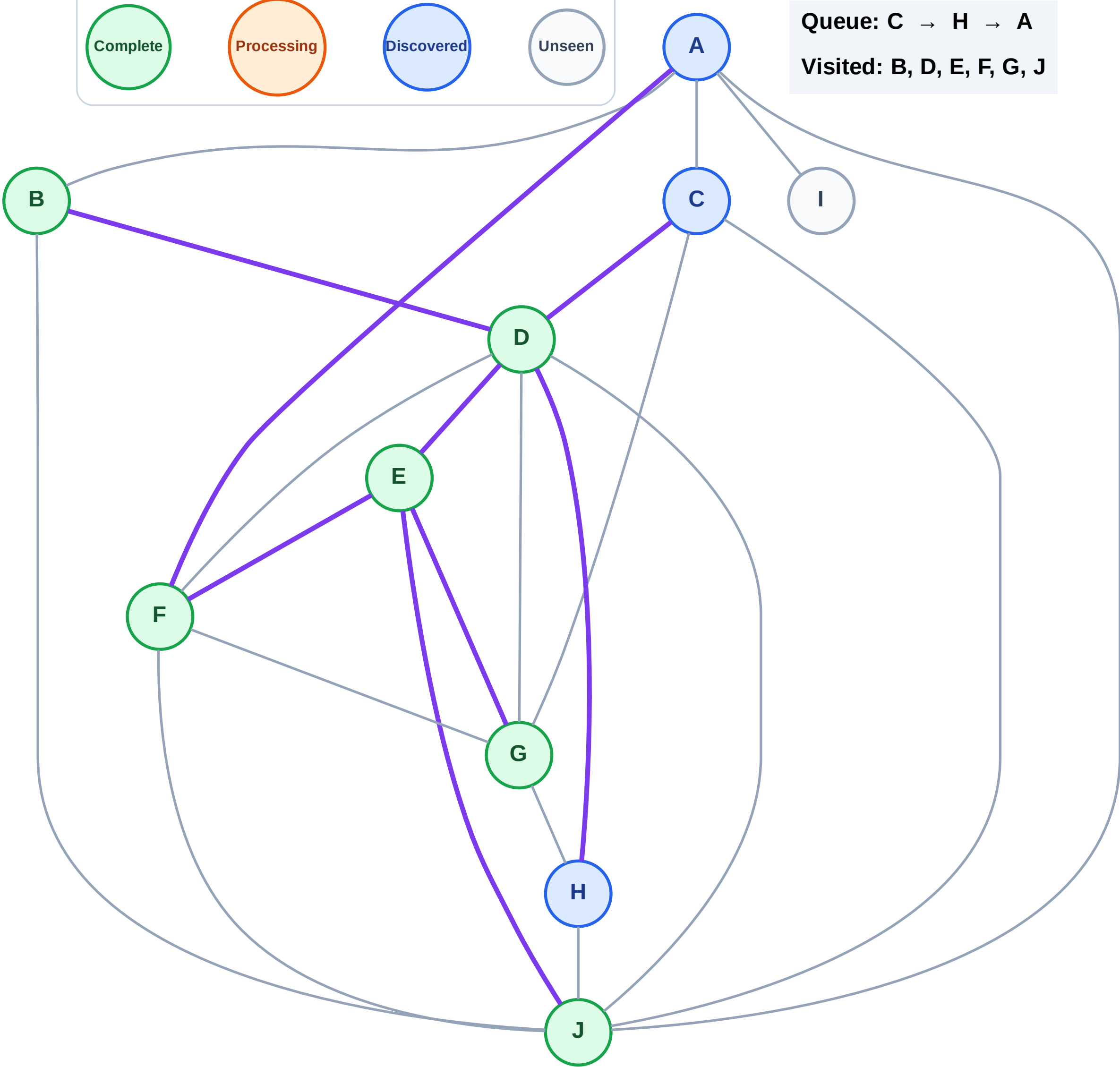
Step 13: No new neighbors from B

Legend



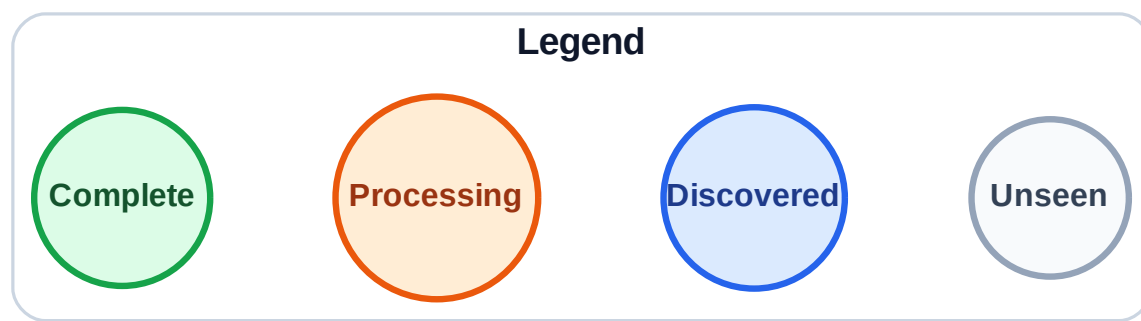
Queue: C → H → A

Visited: B, D, E, F, G, J

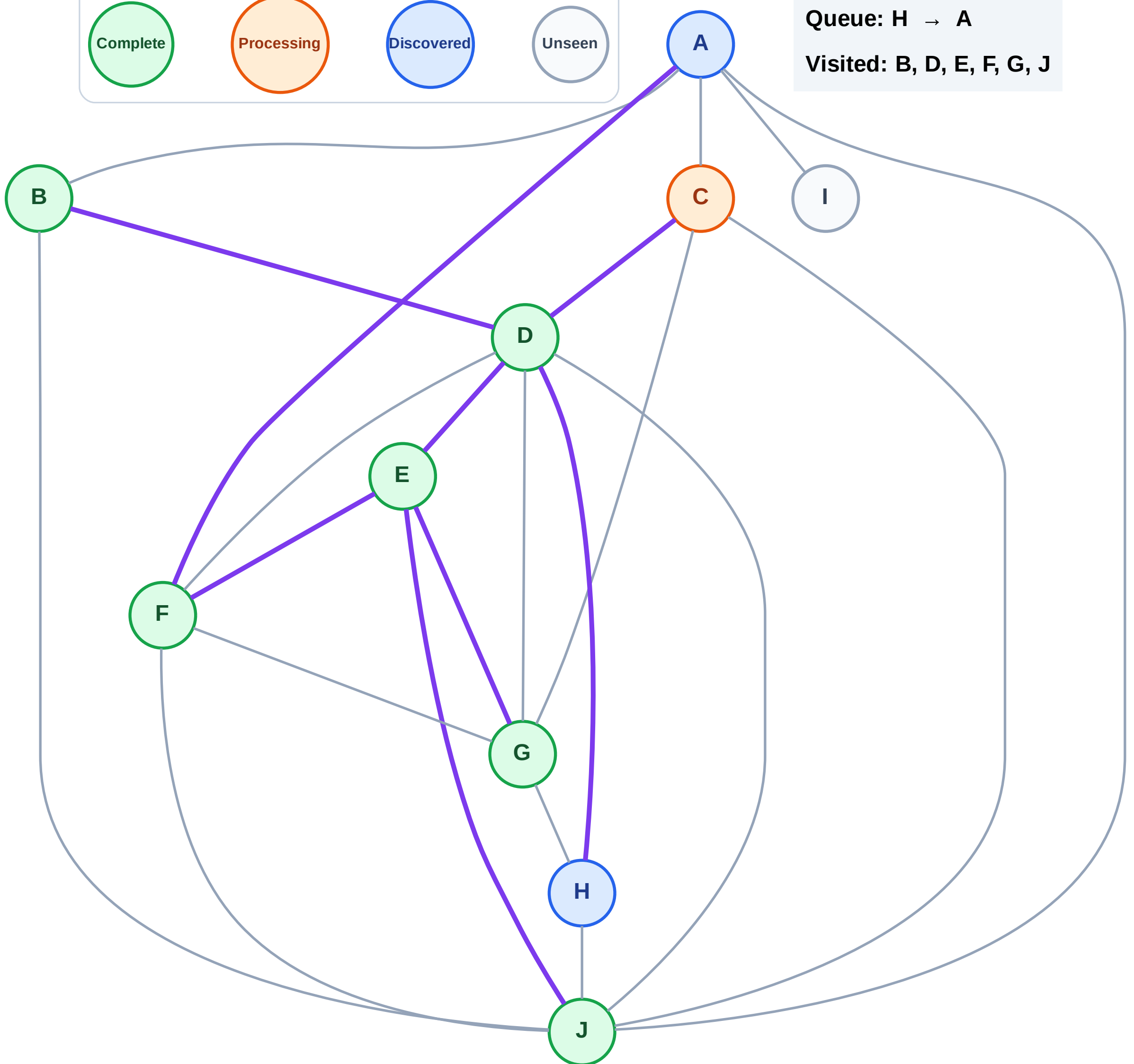


BFS Traversal

Step 14: Process node C



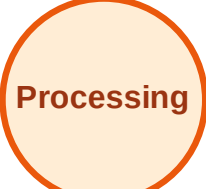
Queue: H → A
Visited: B, D, E, F, G, J



BFS Traversal

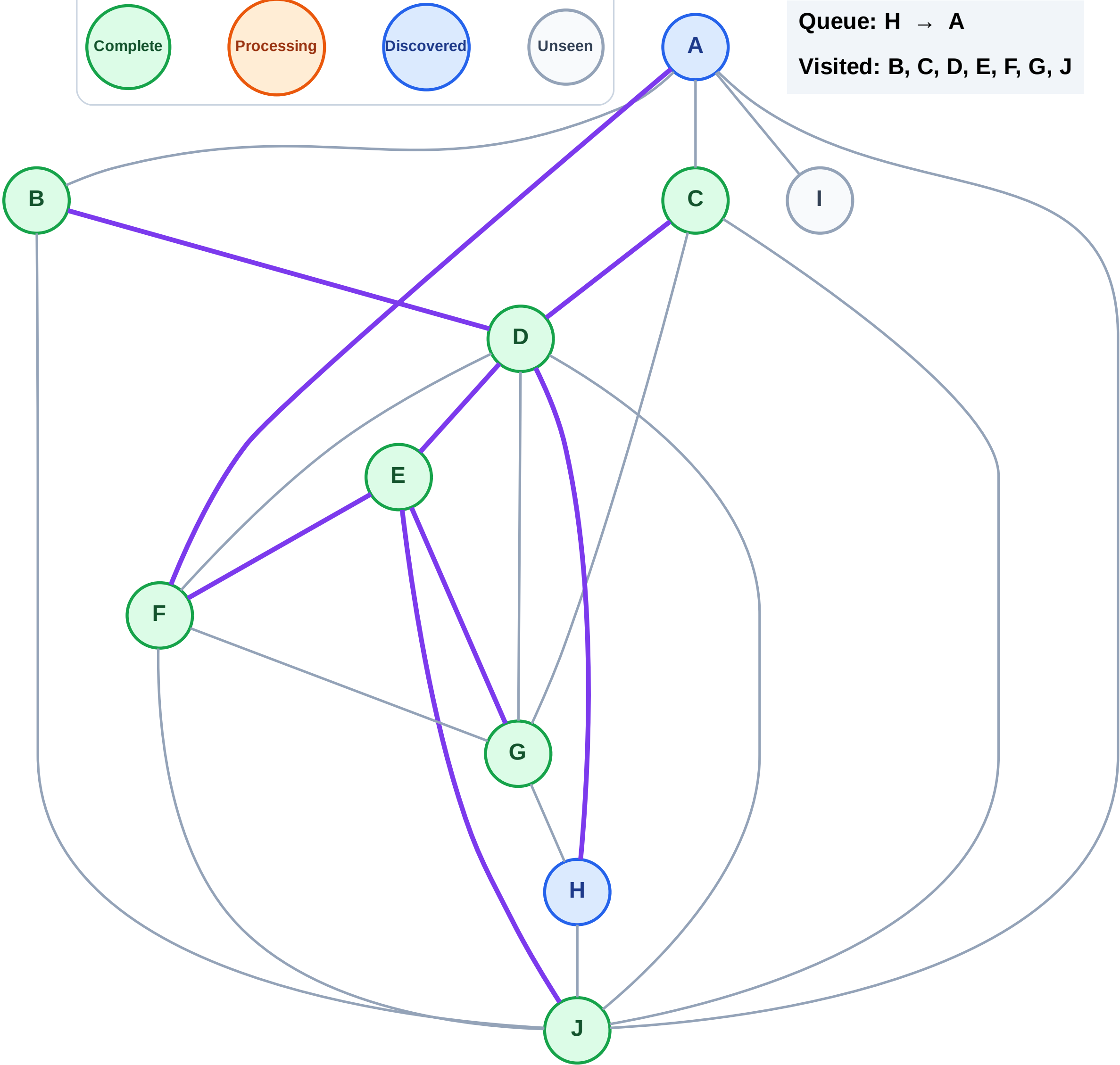
Step 15: No new neighbors from C

Legend



Queue: H → A

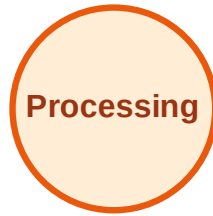
Visited: B, C, D, E, F, G, J



BFS Traversal

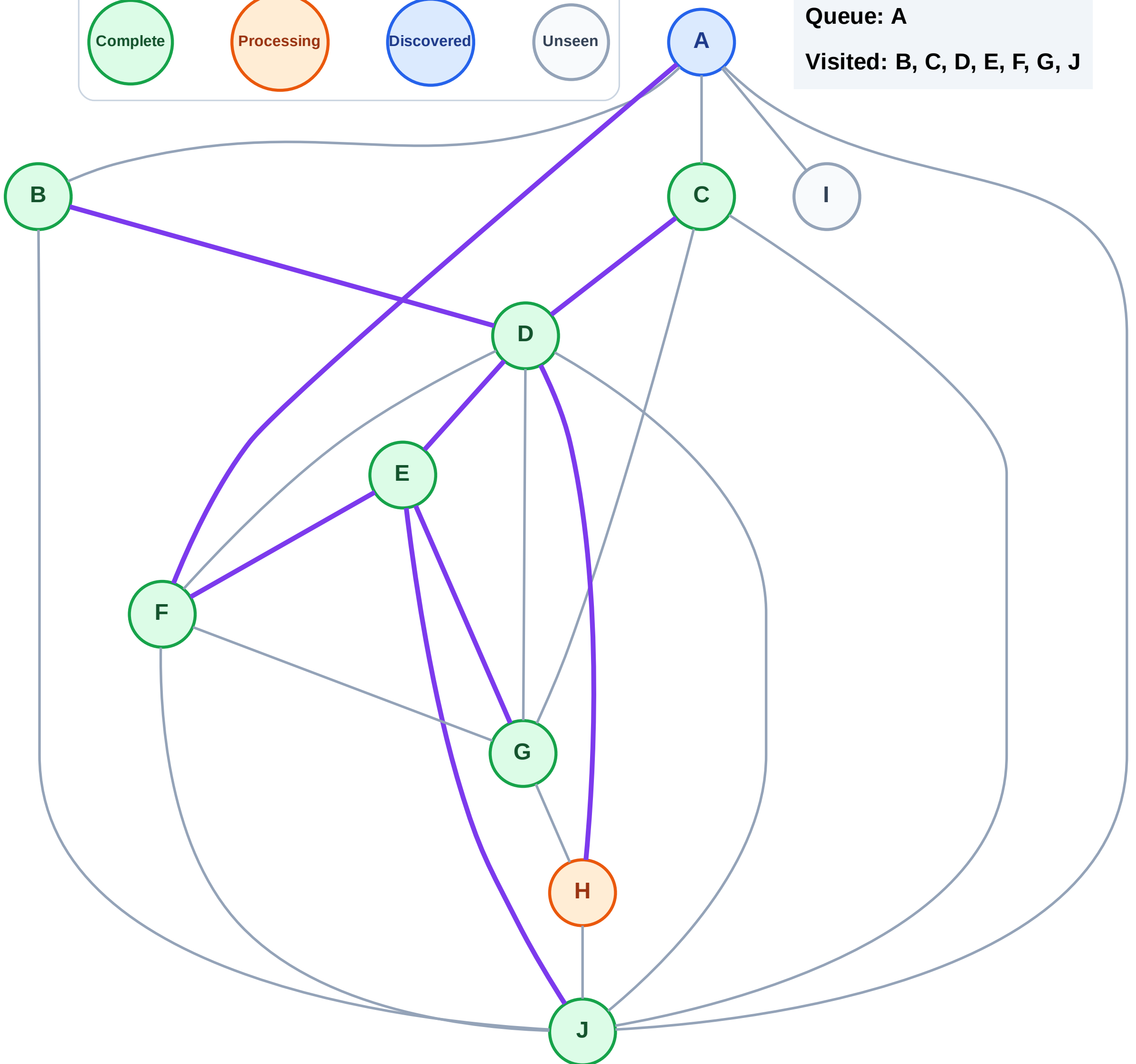
Step 16: Process node H

Legend



Queue: A

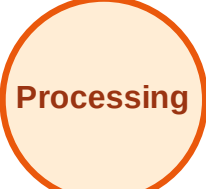
Visited: B, C, D, E, F, G, J



BFS Traversal

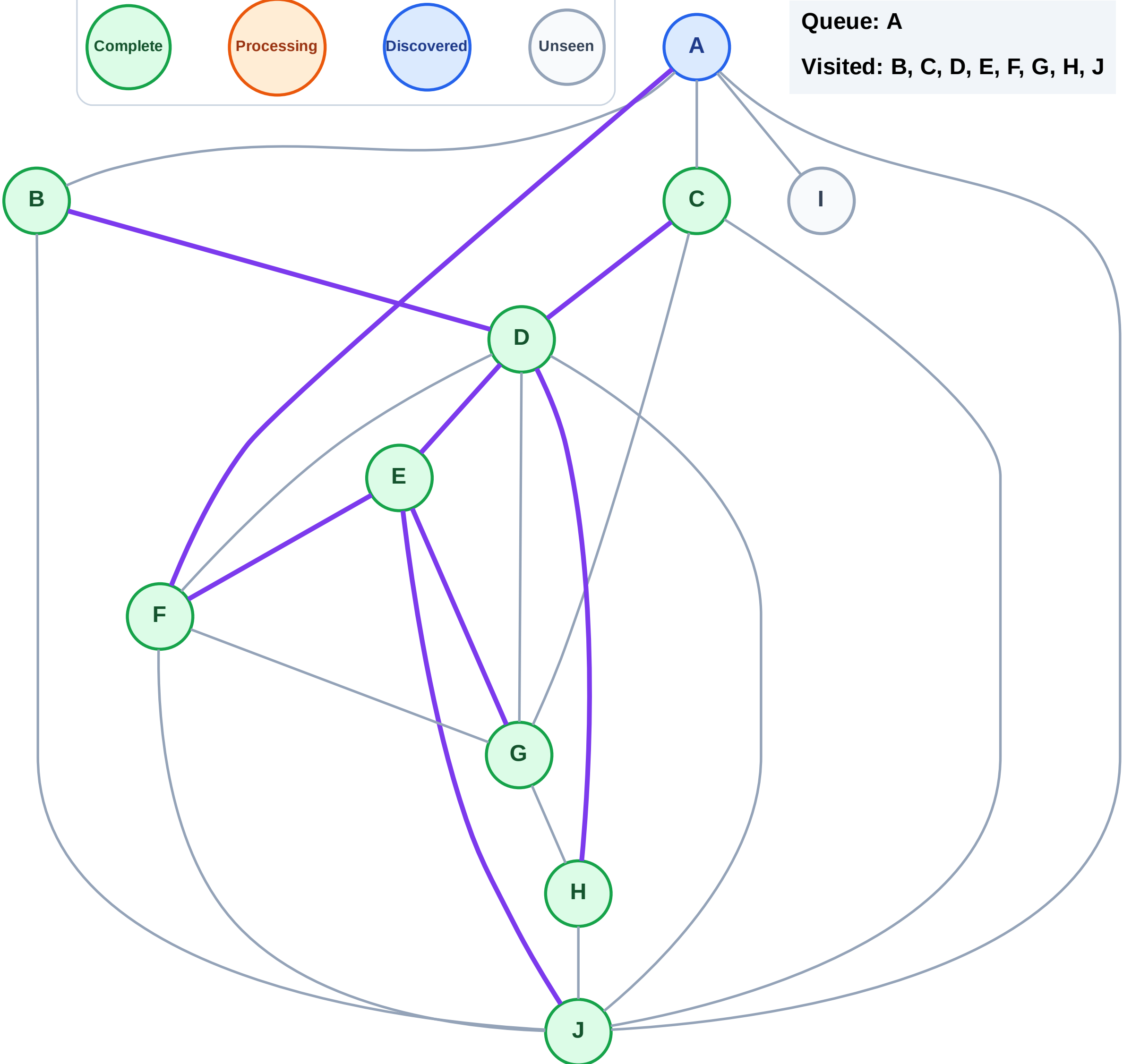
Step 17: No new neighbors from H

Legend



Queue: A

Visited: B, C, D, E, F, G, H, J



BFS Traversal

Step 18: Process node A

Legend

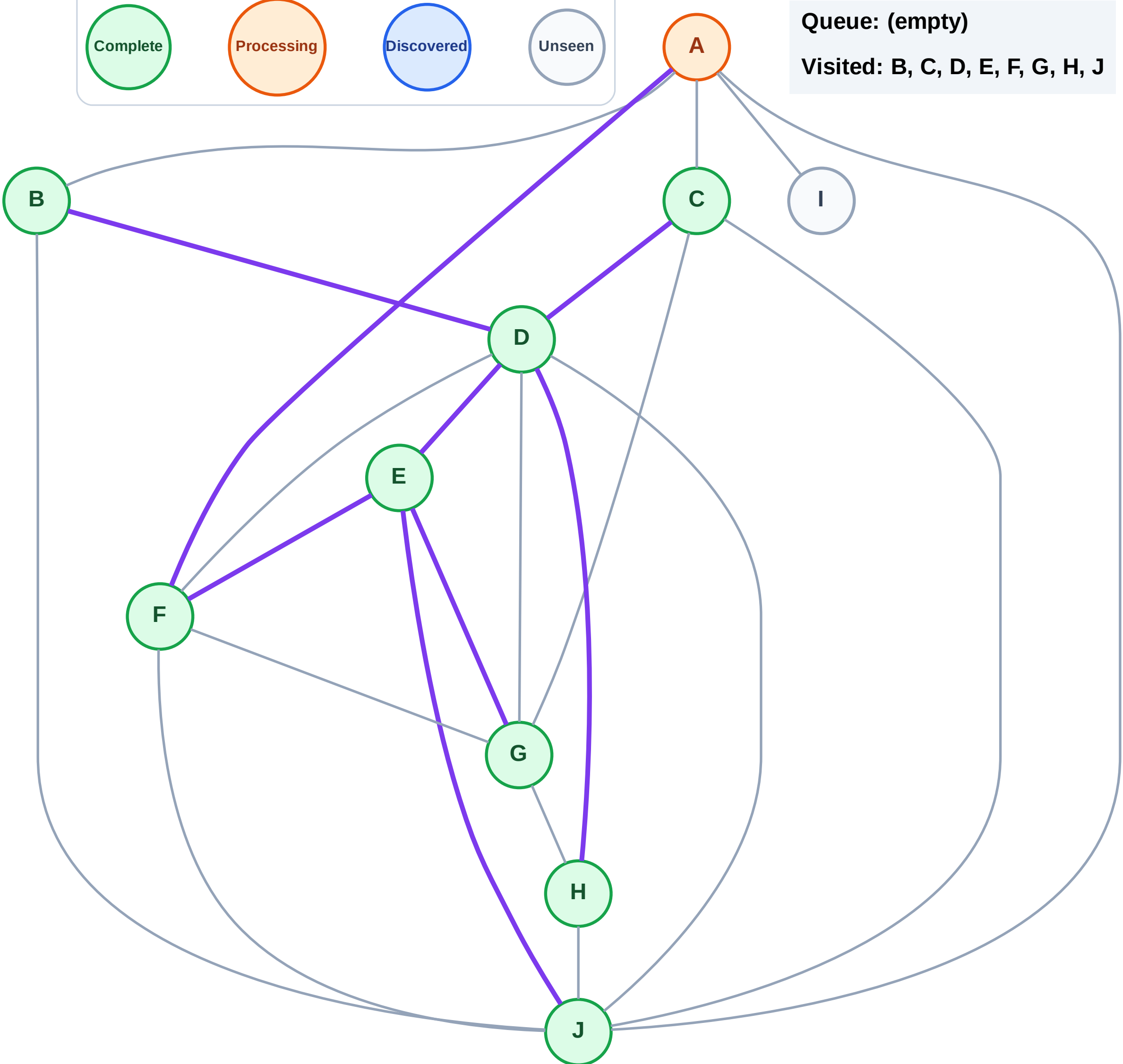
Complete

Processing

Discovered

Unseen

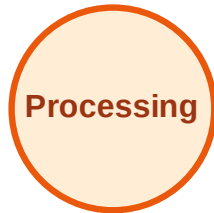
Queue: (empty)
Visited: B, C, D, E, F, G, H, J



BFS Traversal

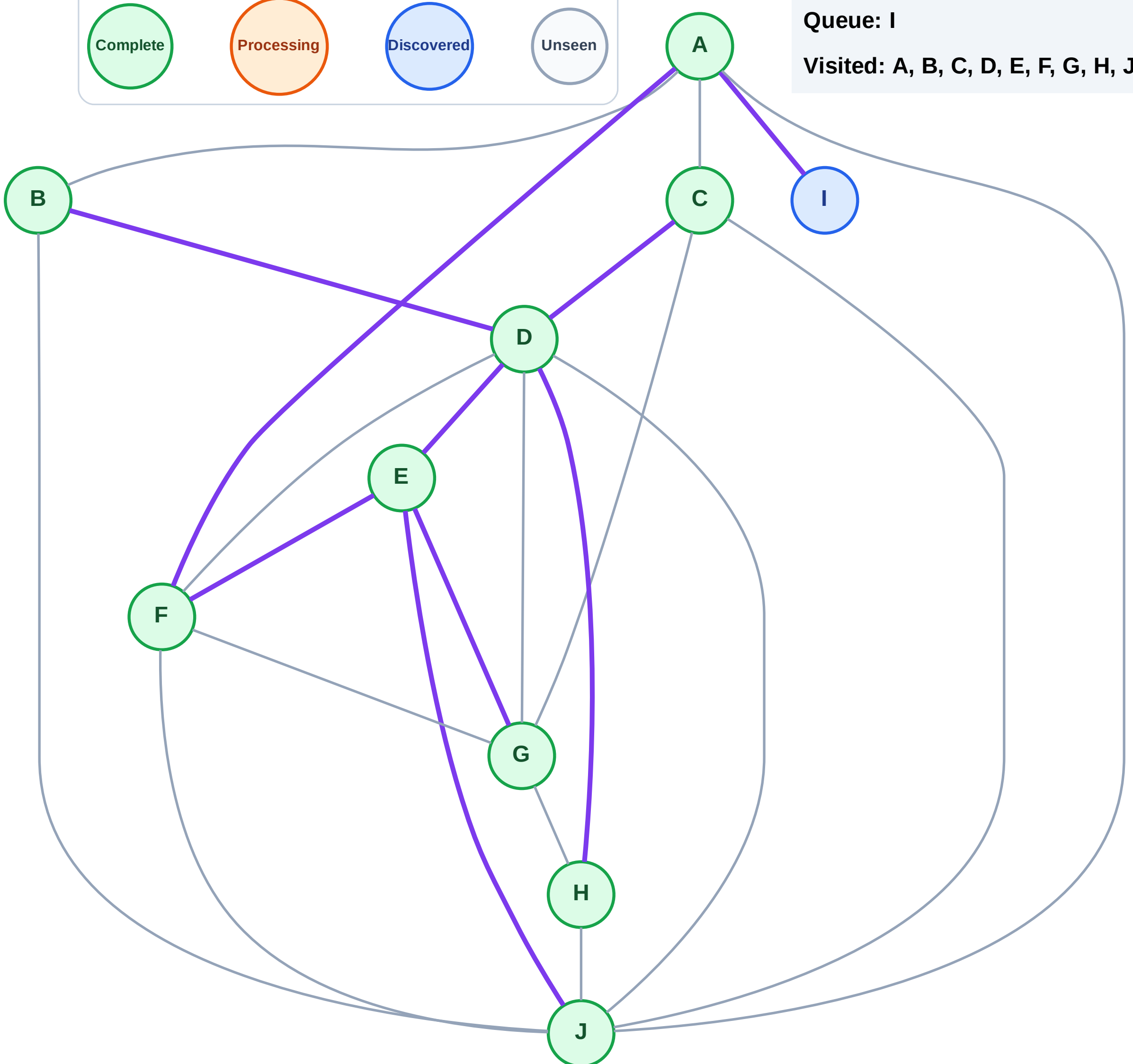
Step 19: Discover I from A

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 20: Process node I

Legend

Complete

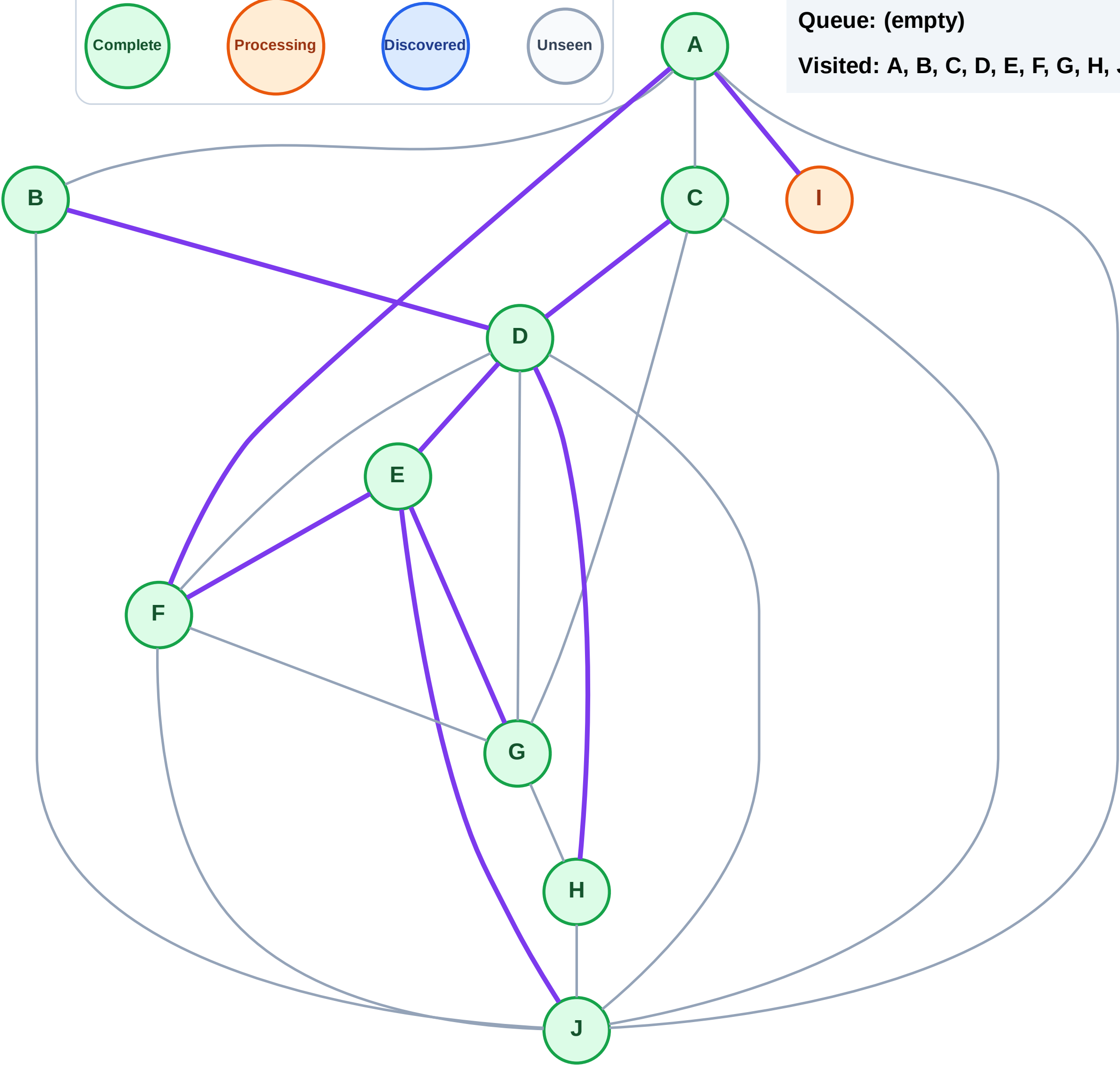
Processing

Discovered

Unseen

Queue: (empty)

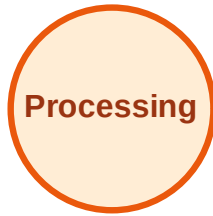
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

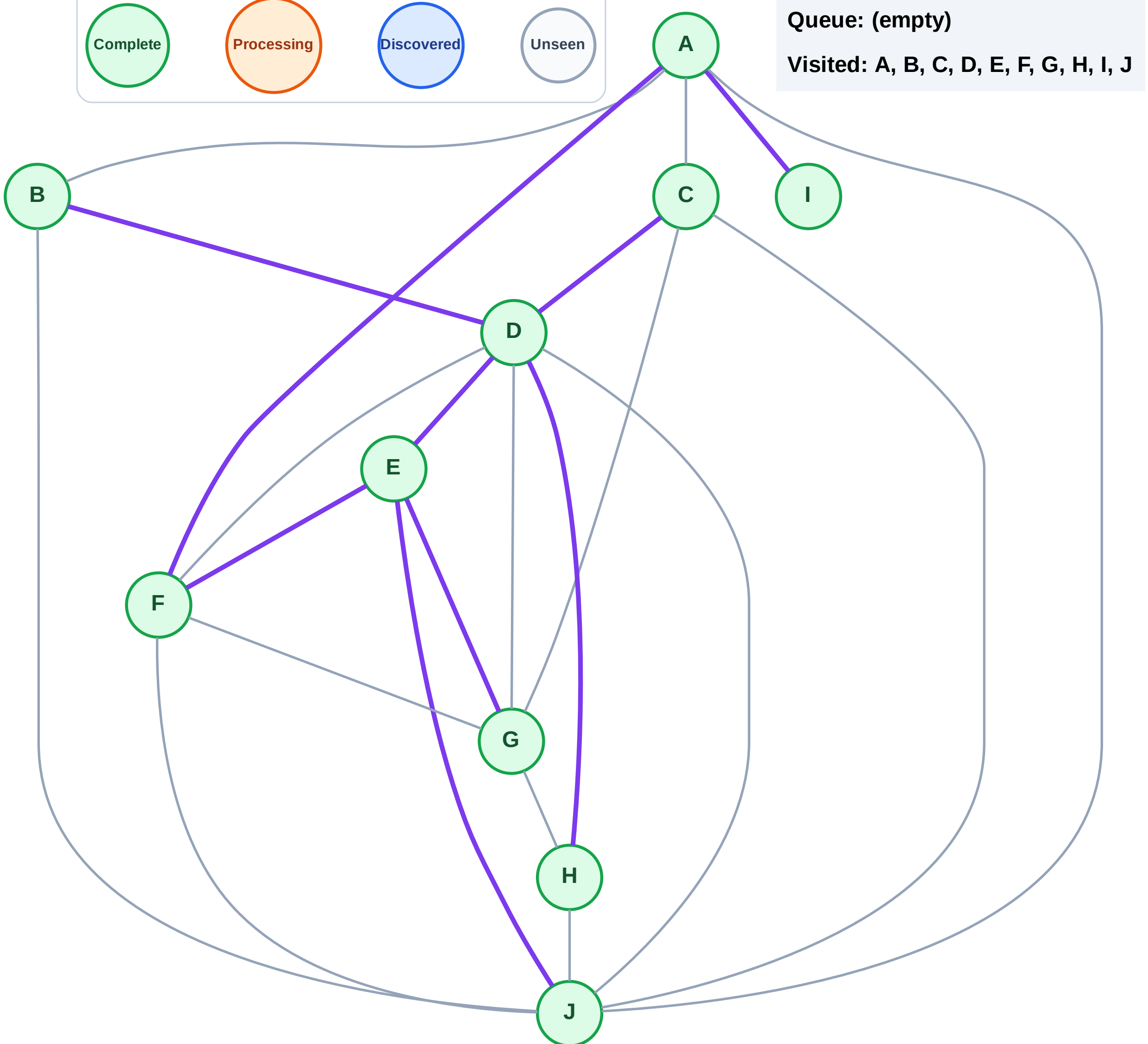
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

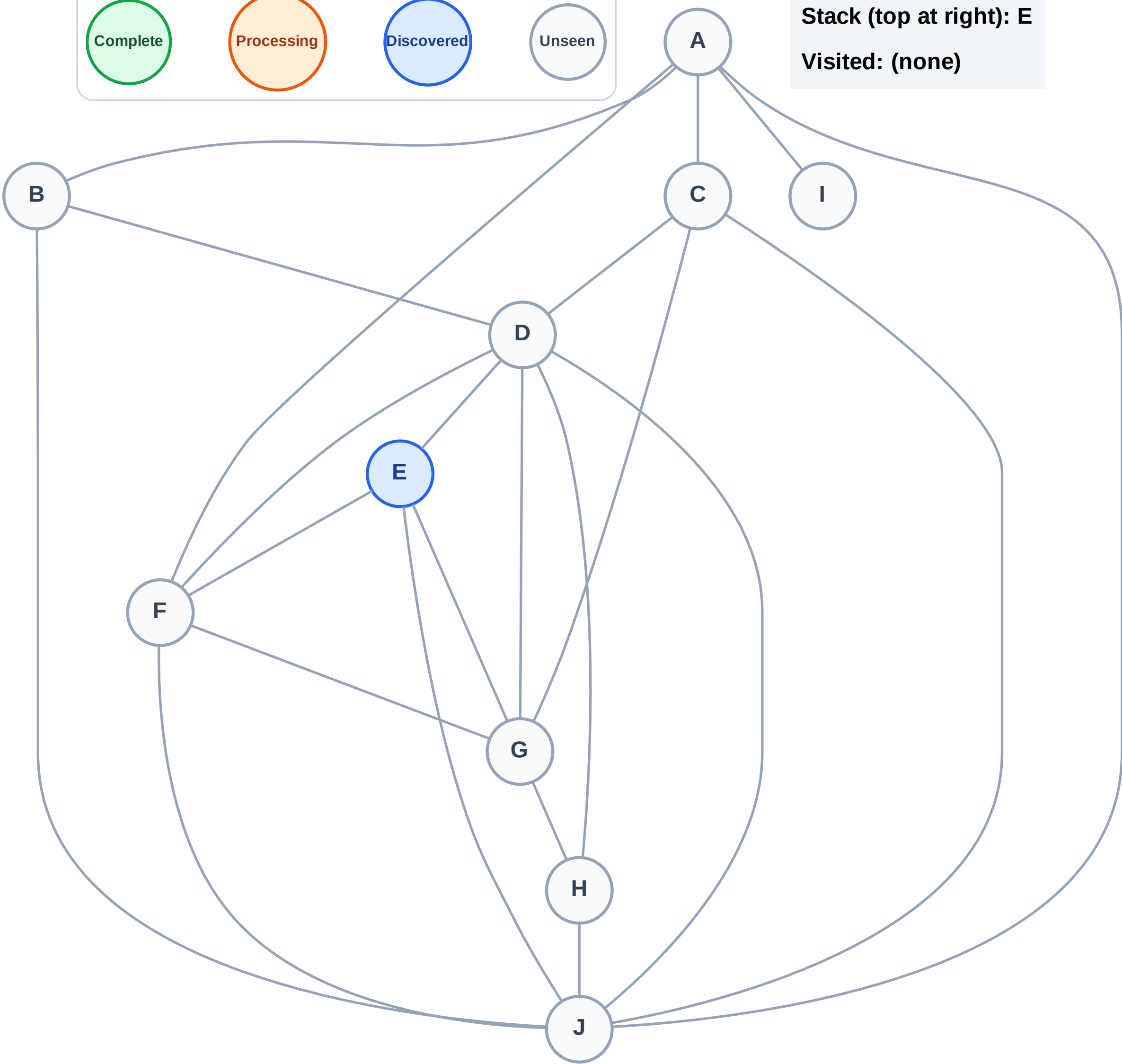
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

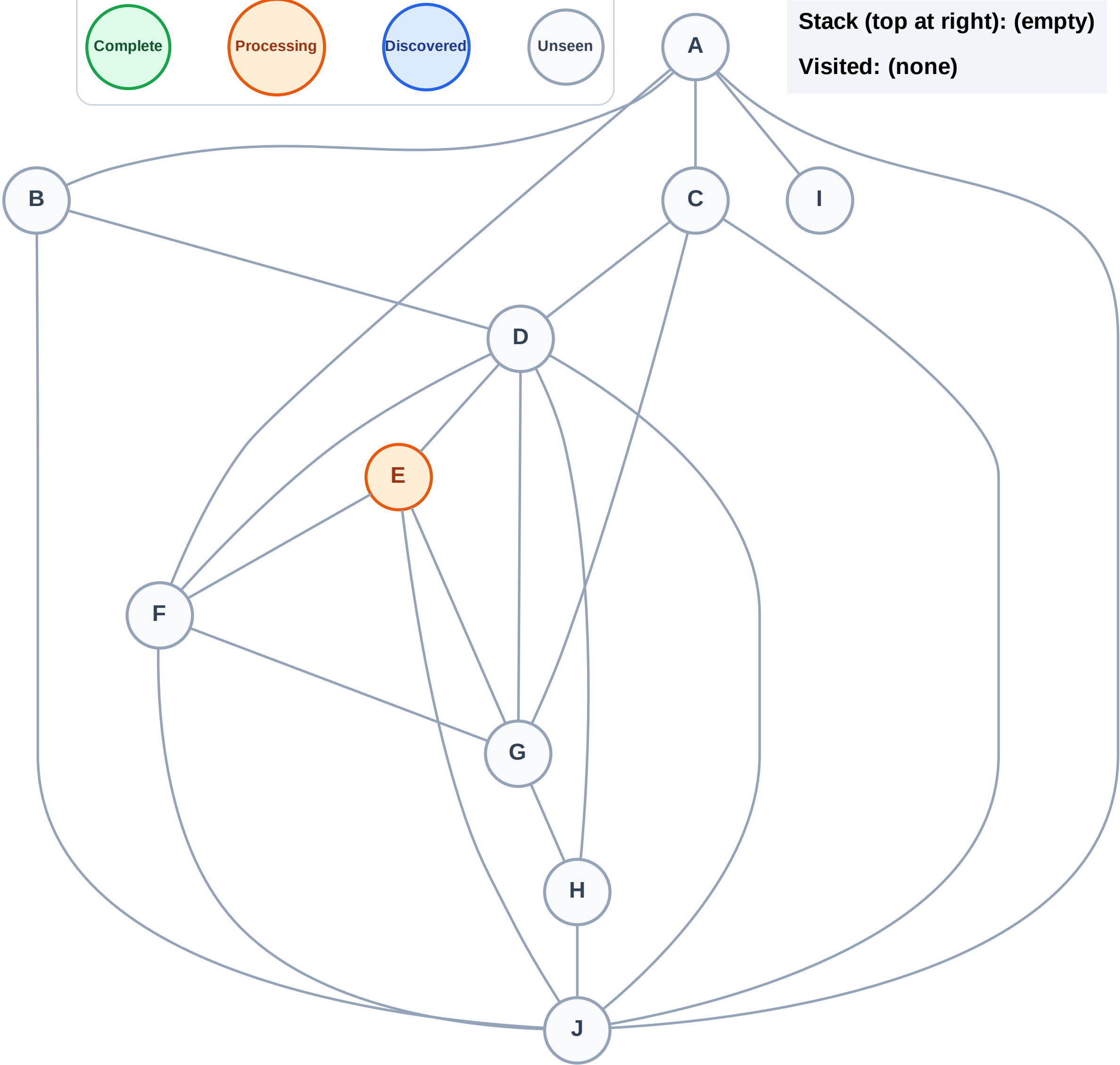
Complete

Processing

Discovered

Unseen

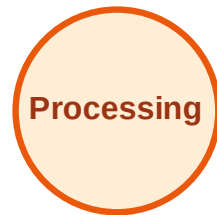
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

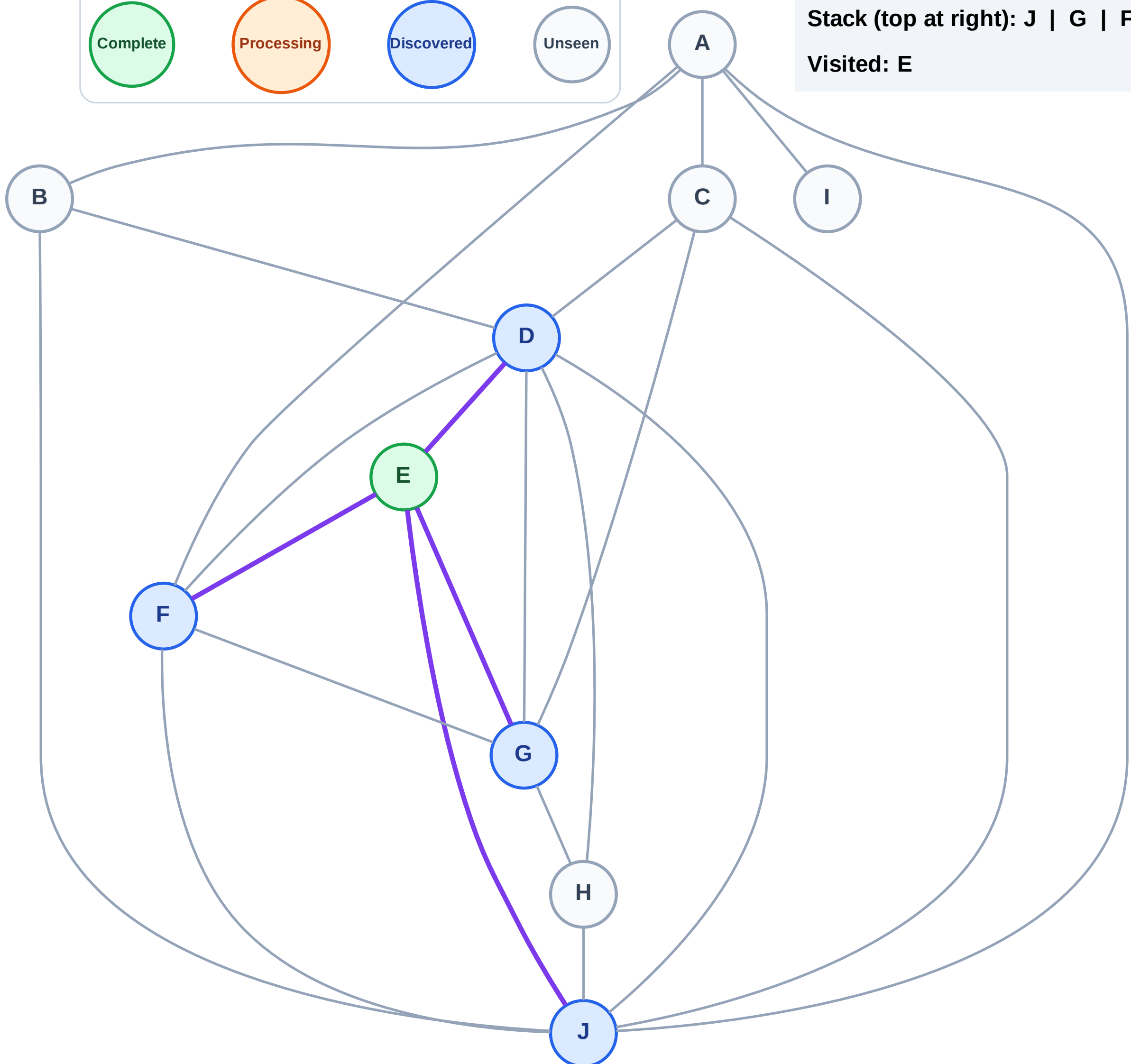
Step 3: Discover J, G, F, D from E

Legend



Stack (top at right): J | G | F | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend

Complete

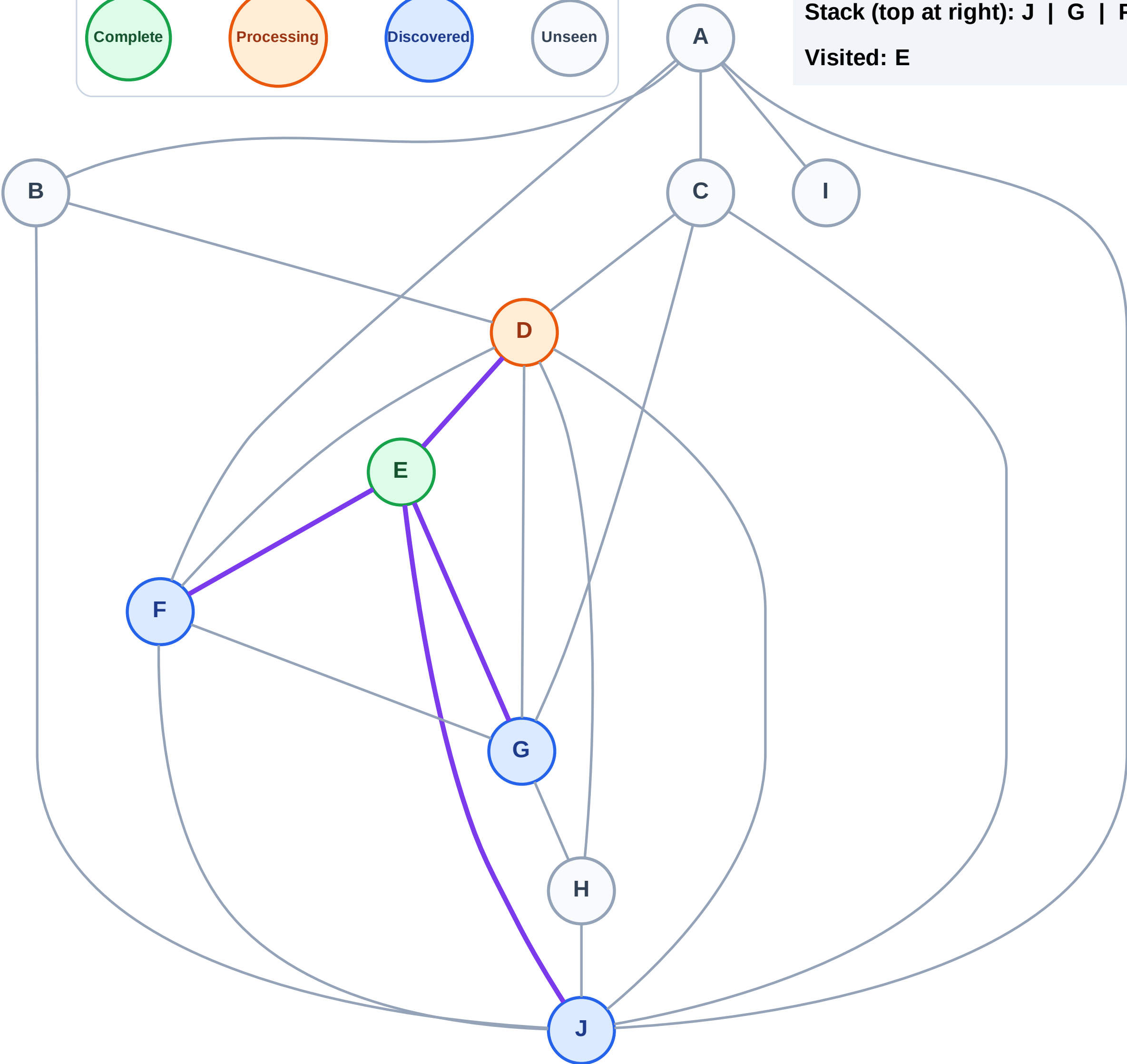
Processing

Discovered

Unseen

Stack (top at right): J | G | F

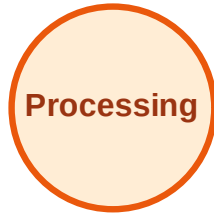
Visited: E



DFS Traversal

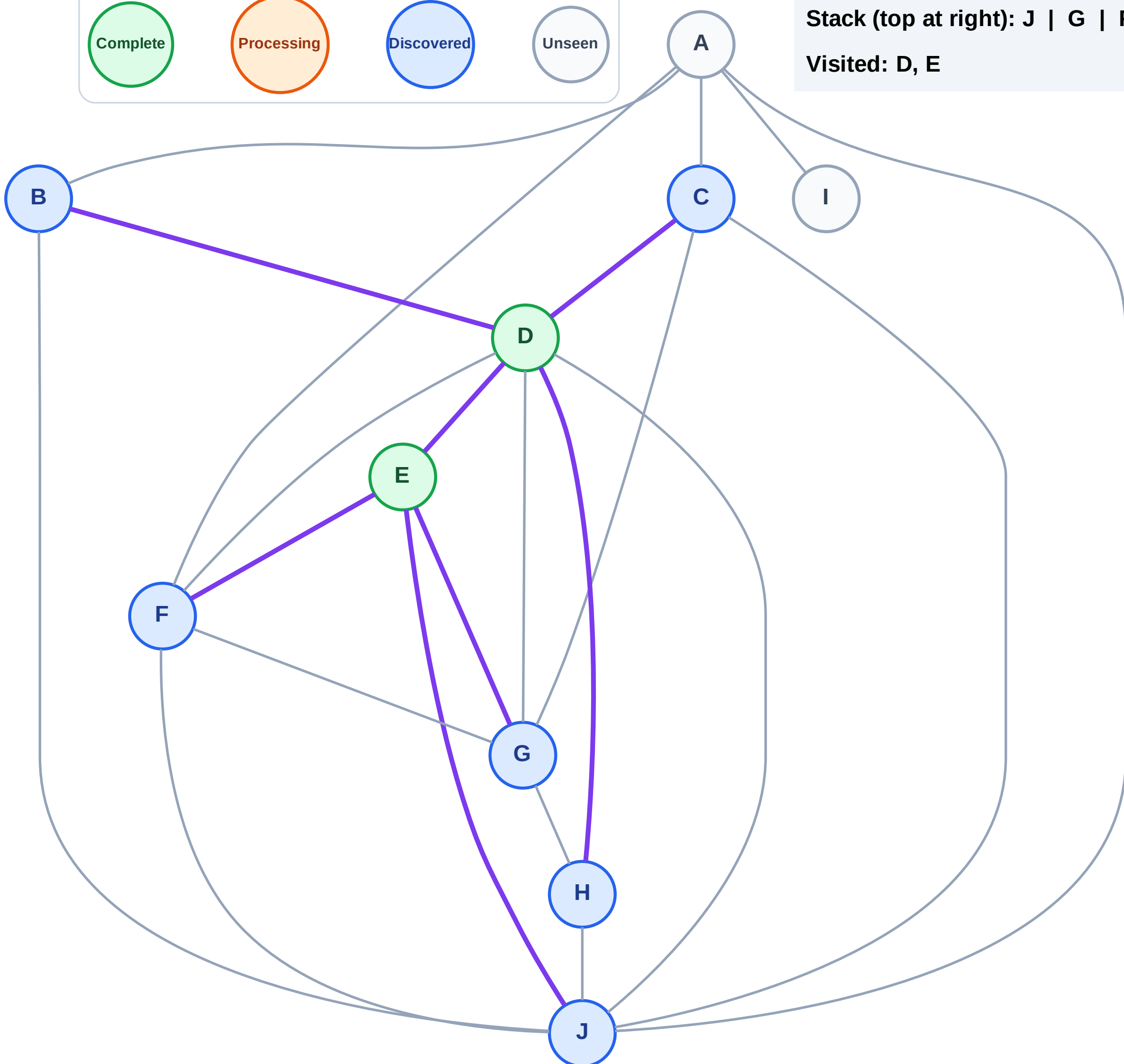
Step 5: Discover H, C, B from D

Legend



Stack (top at right): J | G | F | H | C | B

Visited: D, E



DFS Traversal

Step 6: Process node B

Legend

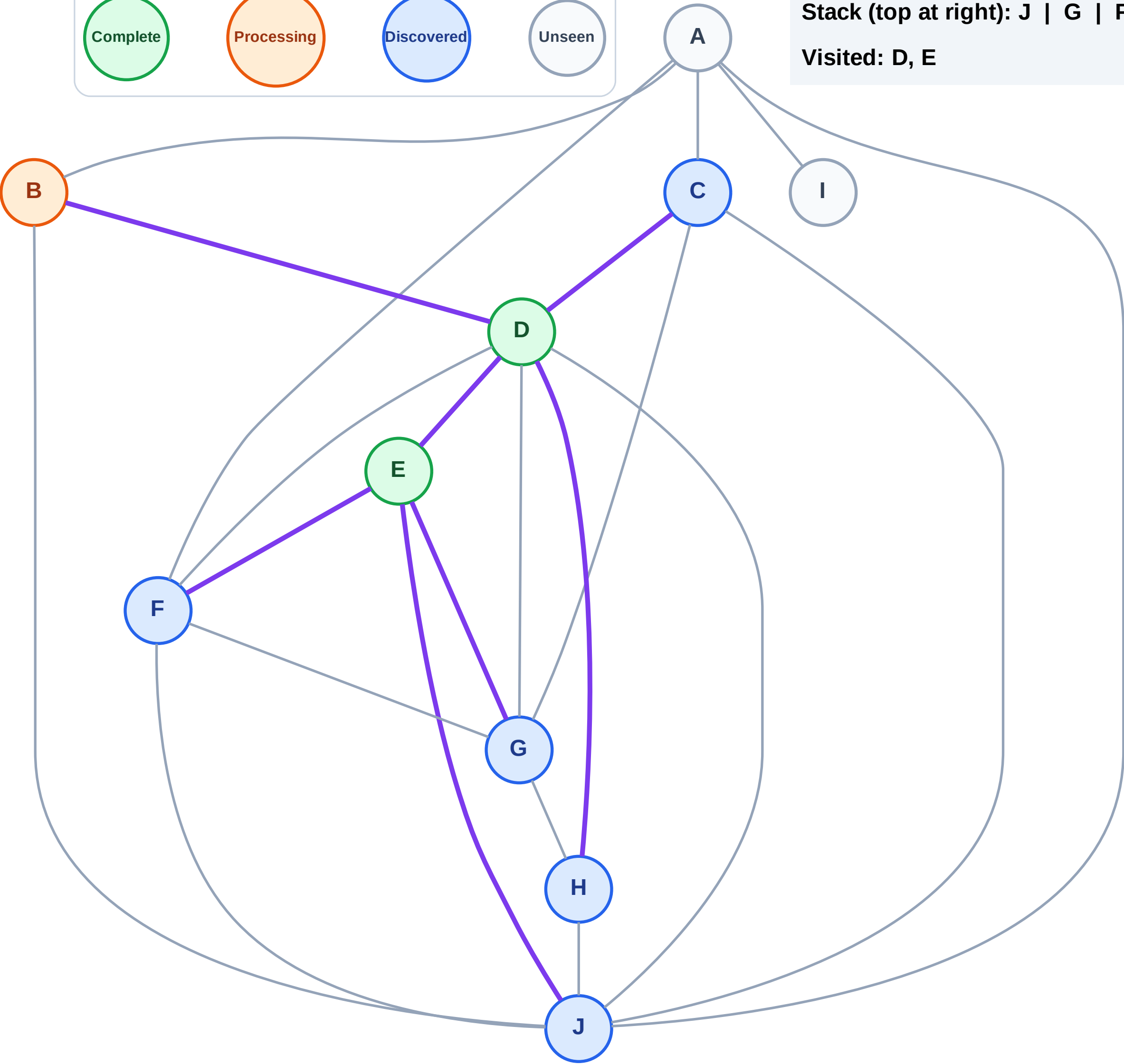
Complete

Processing

Discovered

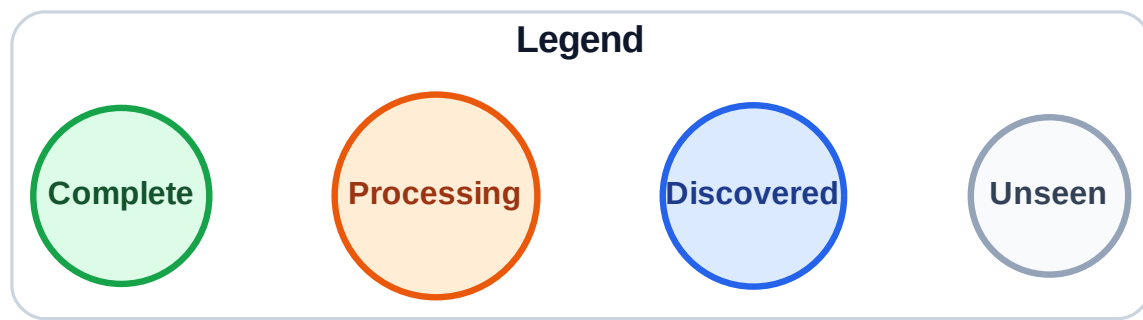
Unseen

Stack (top at right): J | G | F | H | C
Visited: D, E

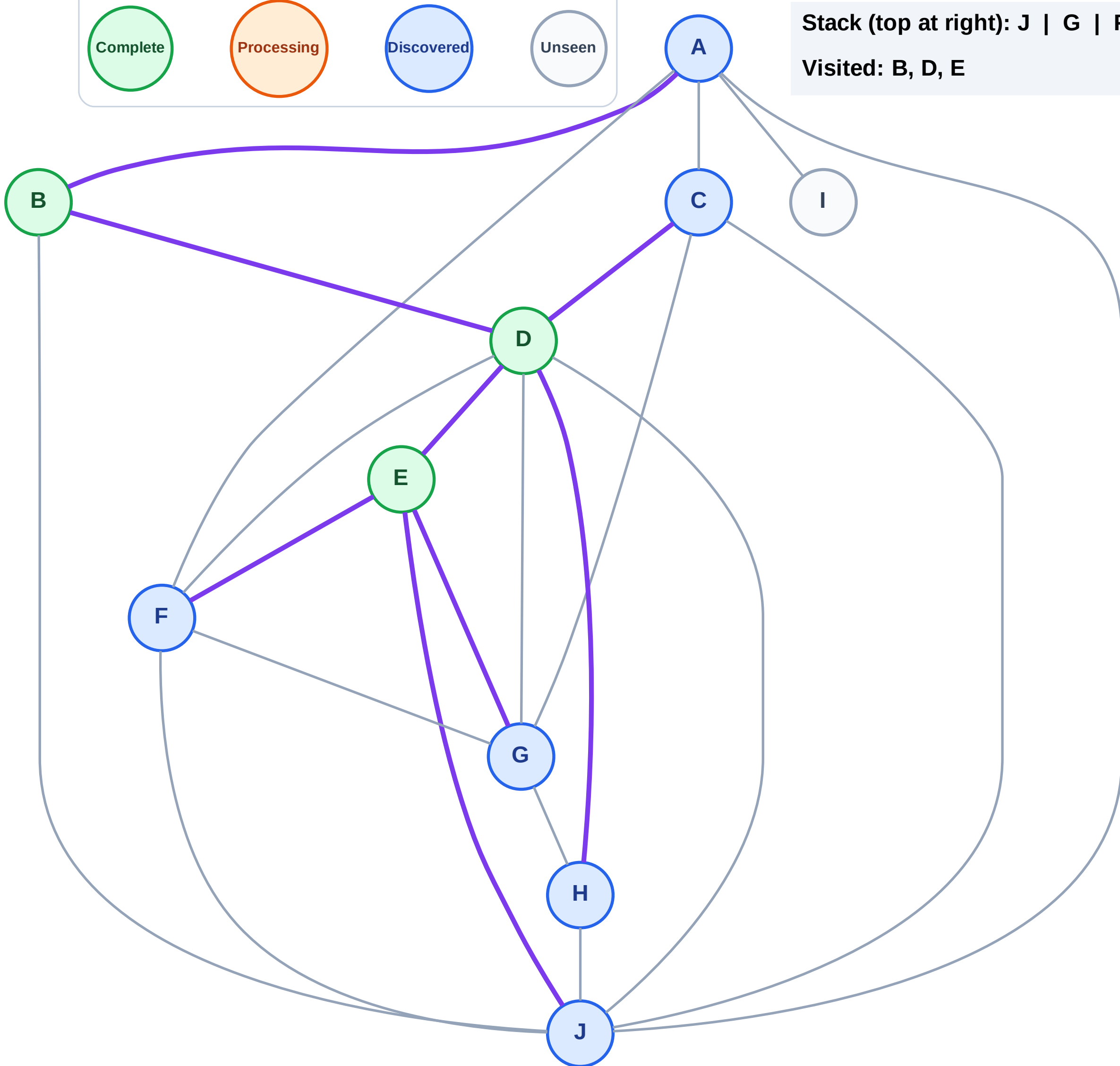


DFS Traversal

Step 7: Discover A from B



Stack (top at right): J | G | F | H | C | A
Visited: B, D, E



DFS Traversal

Step 8: Process node A

Legend

Complete

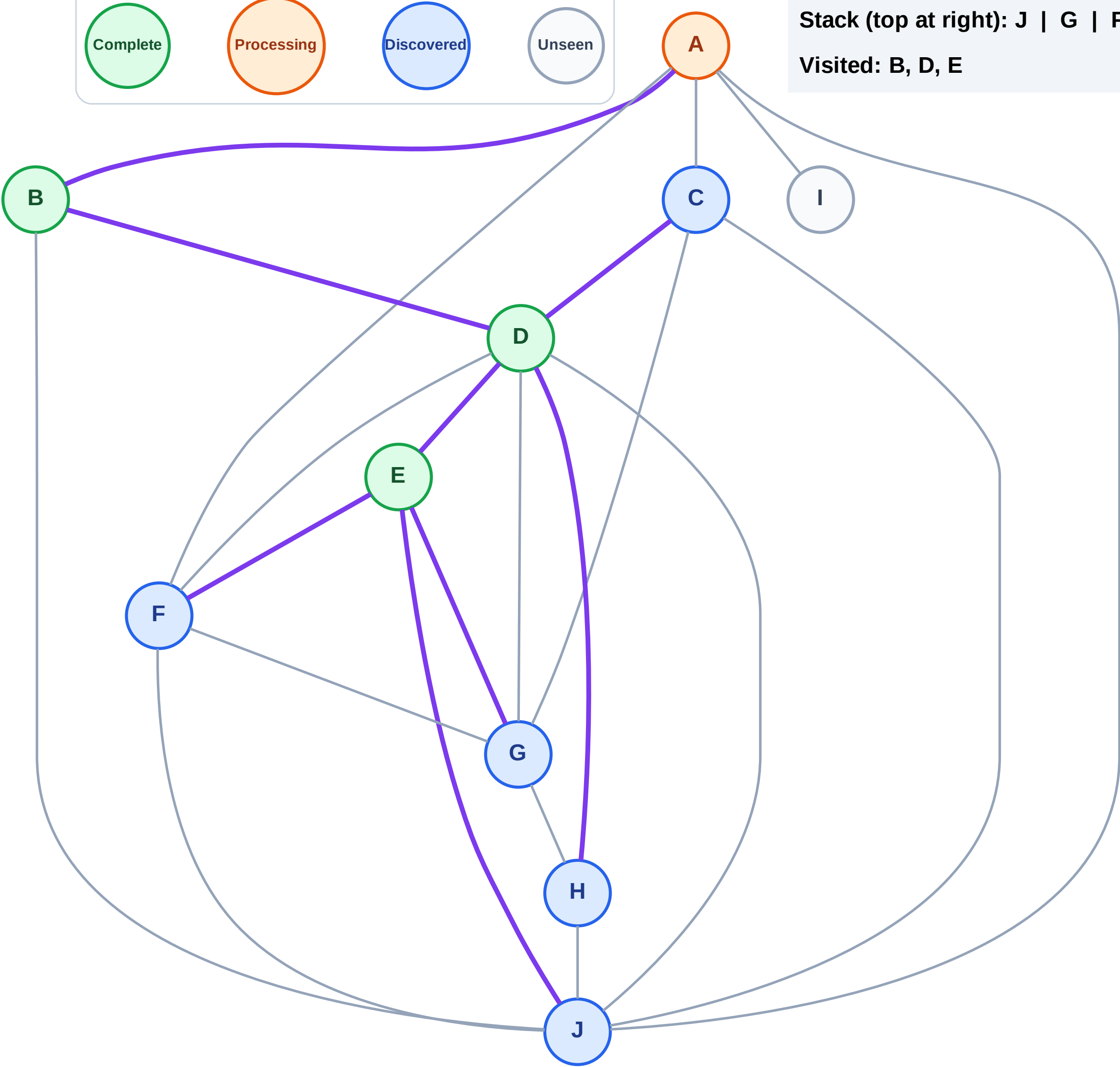
Processing

Discovered

Unseen

Stack (top at right): J | G | F | H | C

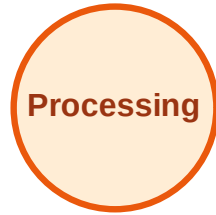
Visited: B, D, E



DFS Traversal

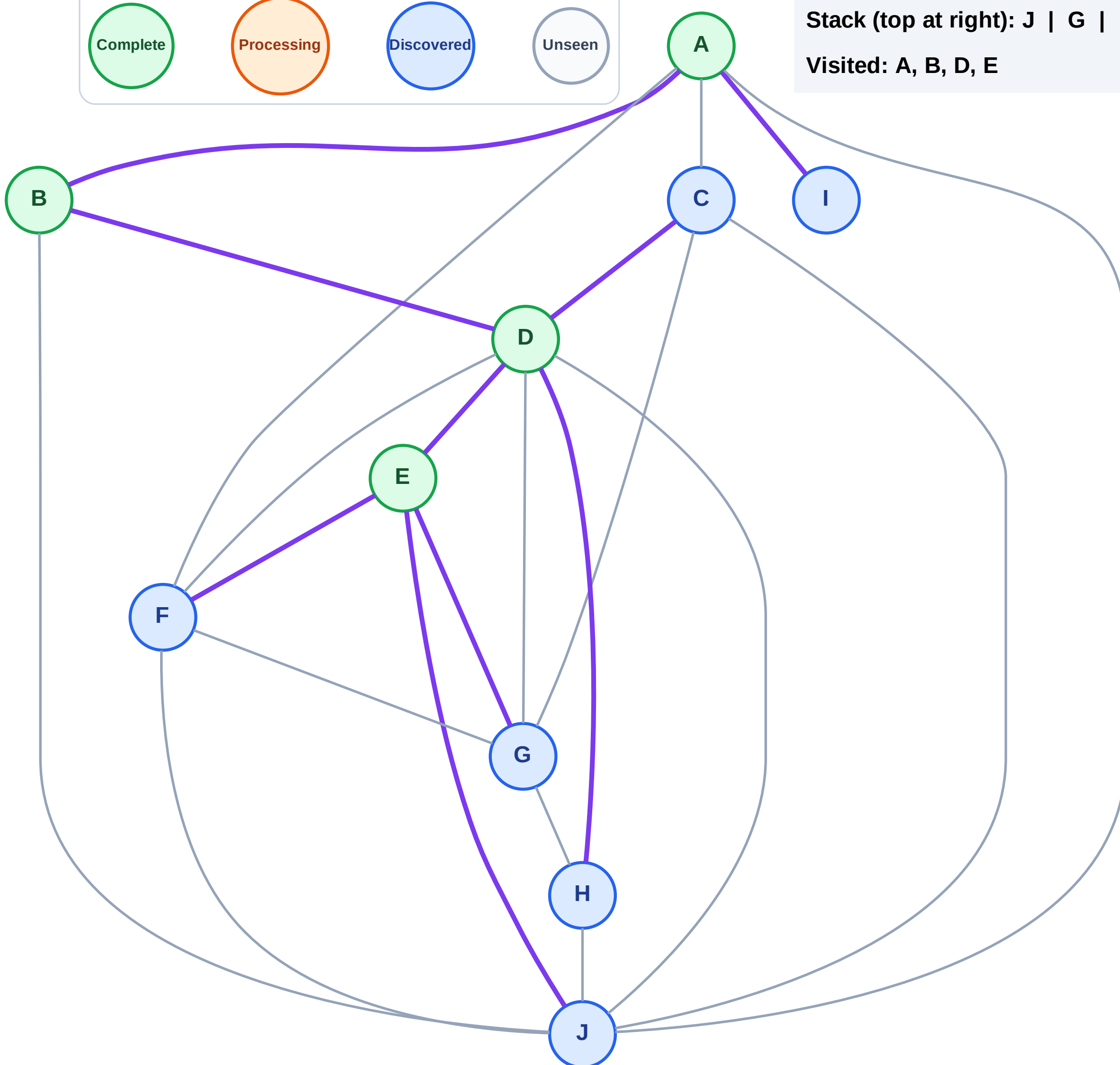
Step 9: Discover I from A

Legend



Stack (top at right): J | G | F | H | C | I

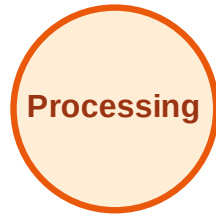
Visited: A, B, D, E



DFS Traversal

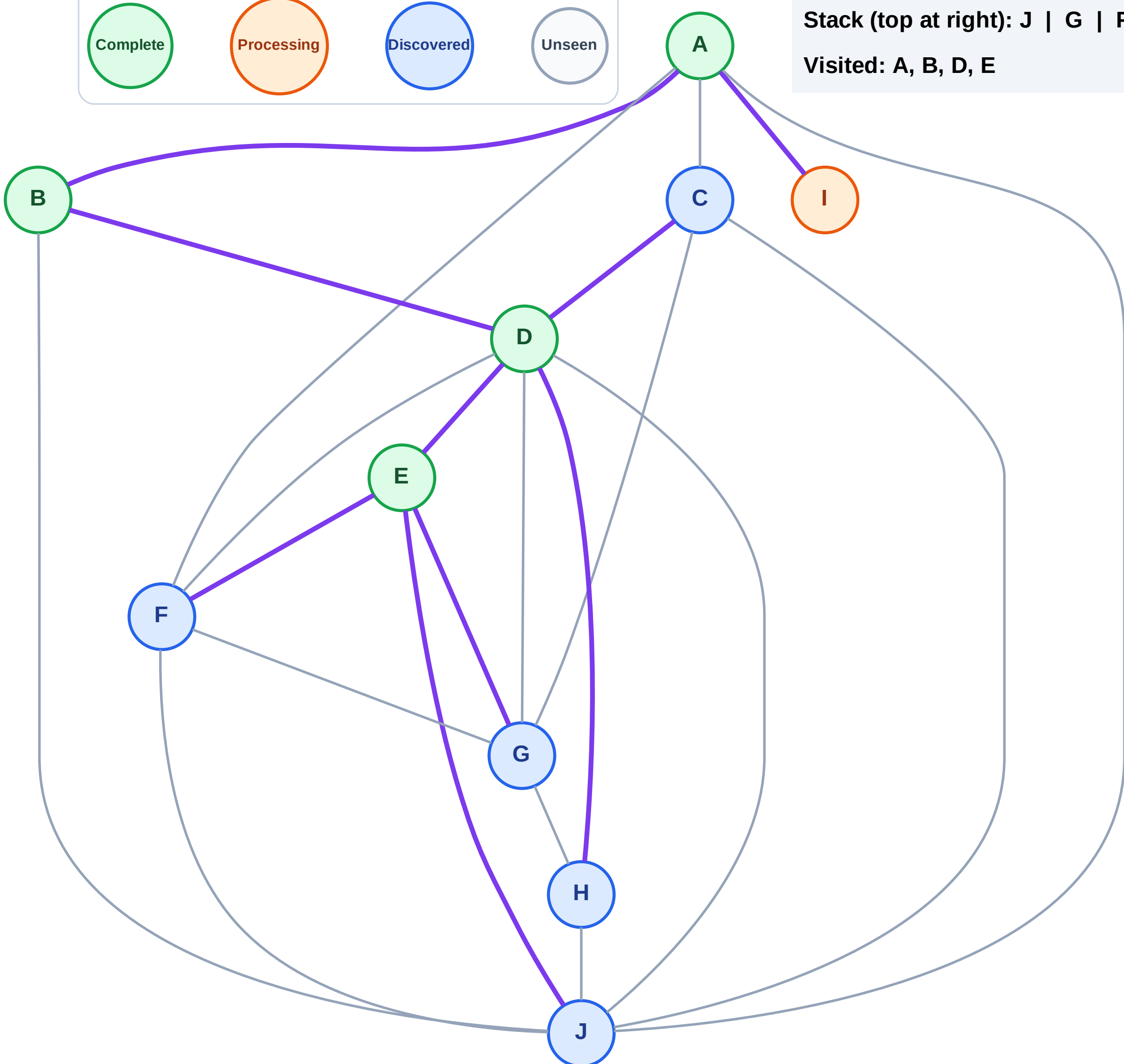
Step 10: Process node I

Legend



Stack (top at right): J | G | F | H | C

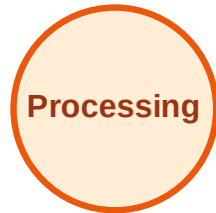
Visited: A, B, D, E



DFS Traversal

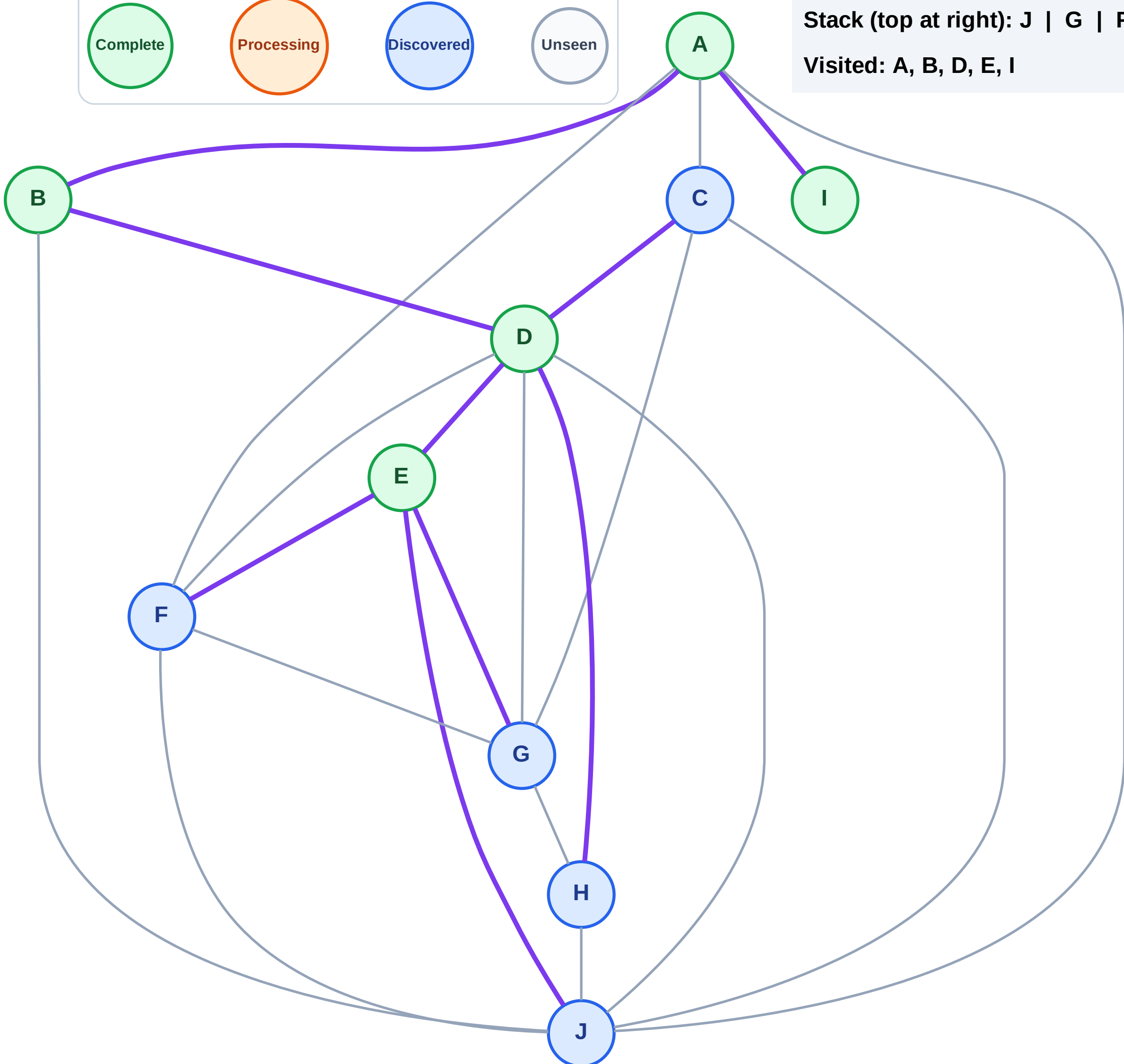
Step 11: No new neighbors from I

Legend



Stack (top at right): J | G | F | H | C

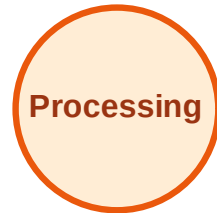
Visited: A, B, D, E, I



DFS Traversal

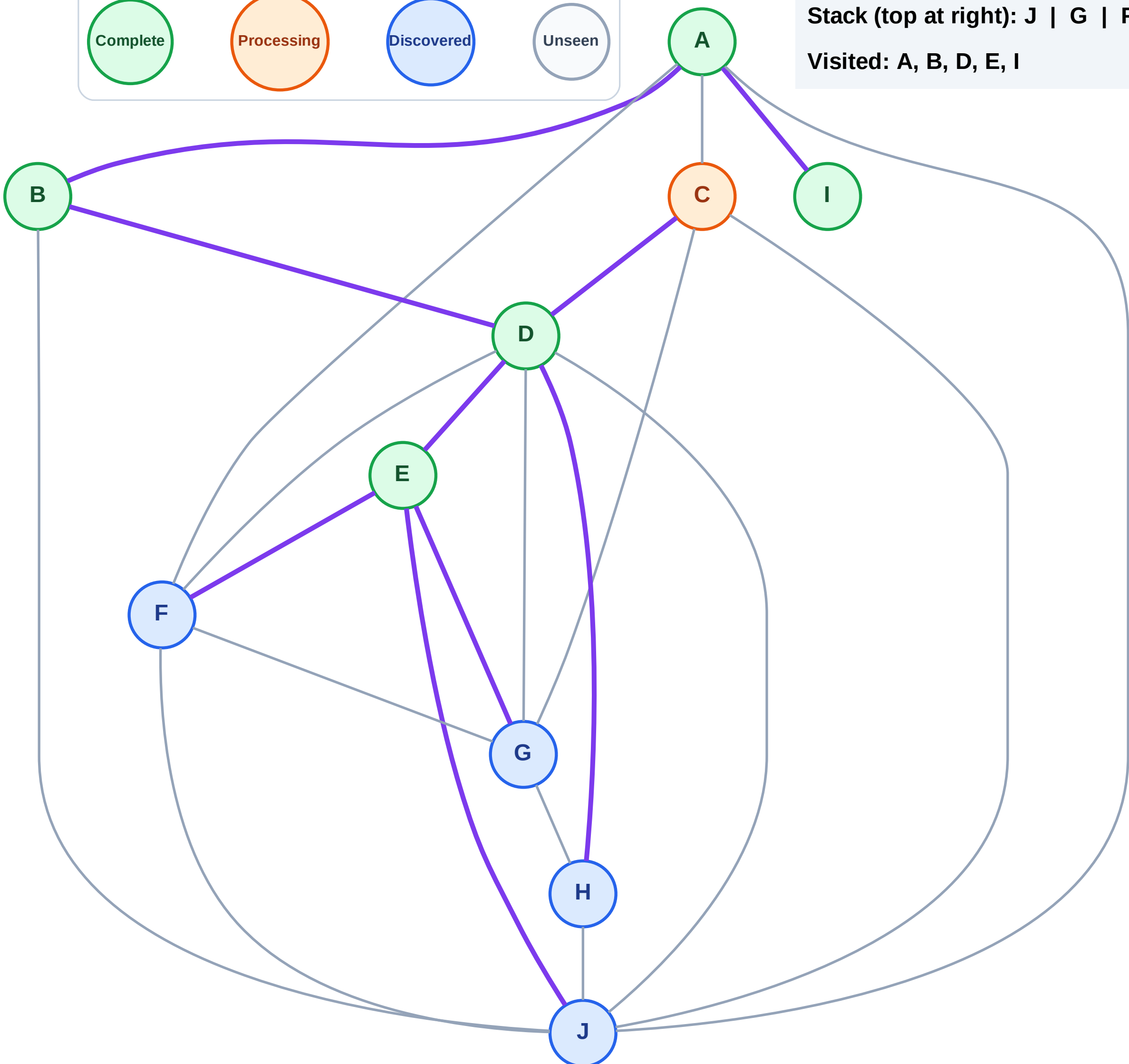
Step 12: Process node C

Legend



Stack (top at right): J | G | F | H

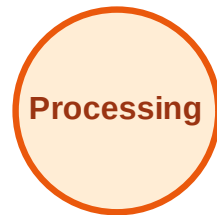
Visited: A, B, D, E, I



DFS Traversal

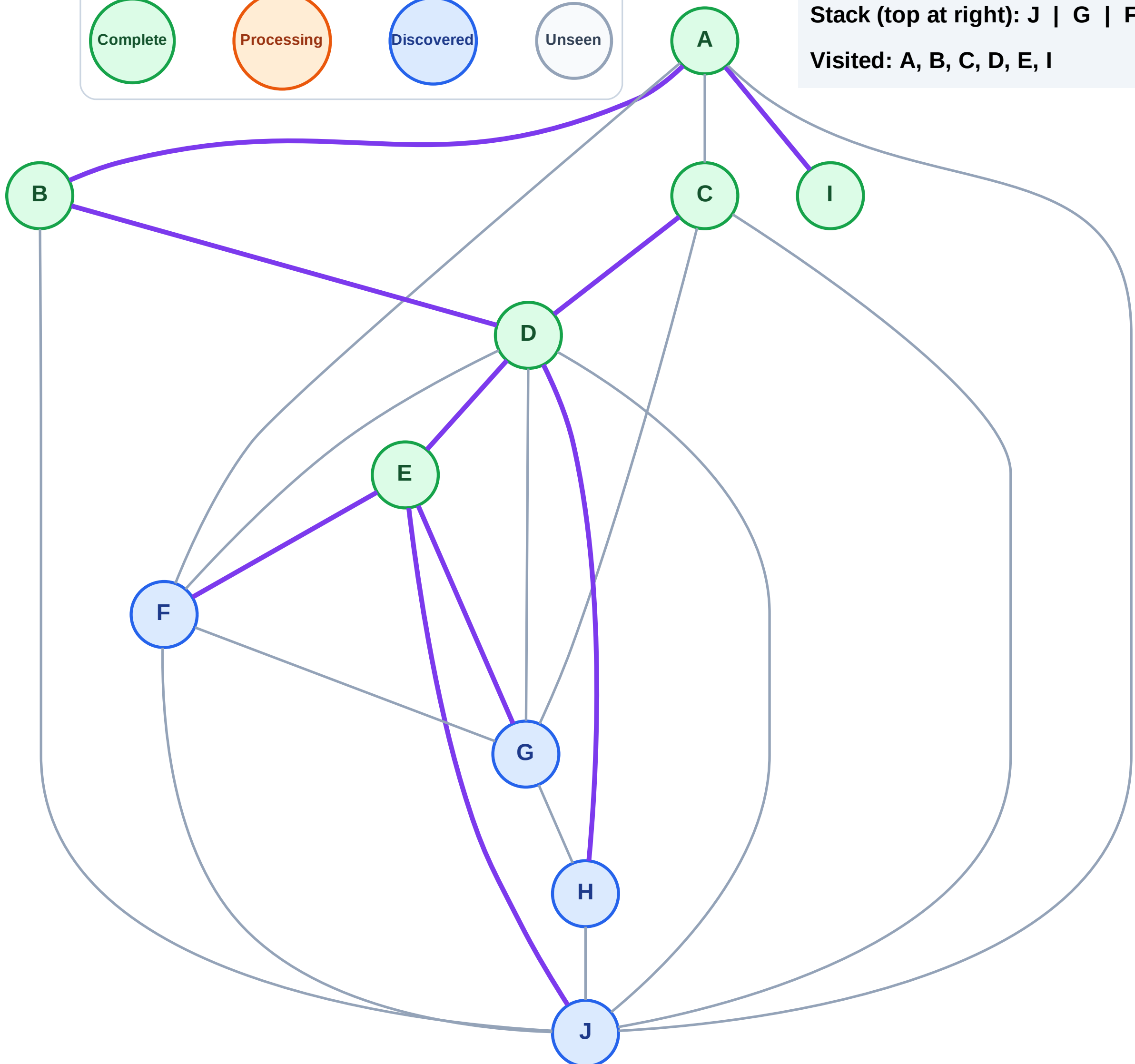
Step 13: No new neighbors from C

Legend



Stack (top at right): J | G | F | H

Visited: A, B, C, D, E, I



DFS Traversal

Step 14: Process node H

Legend

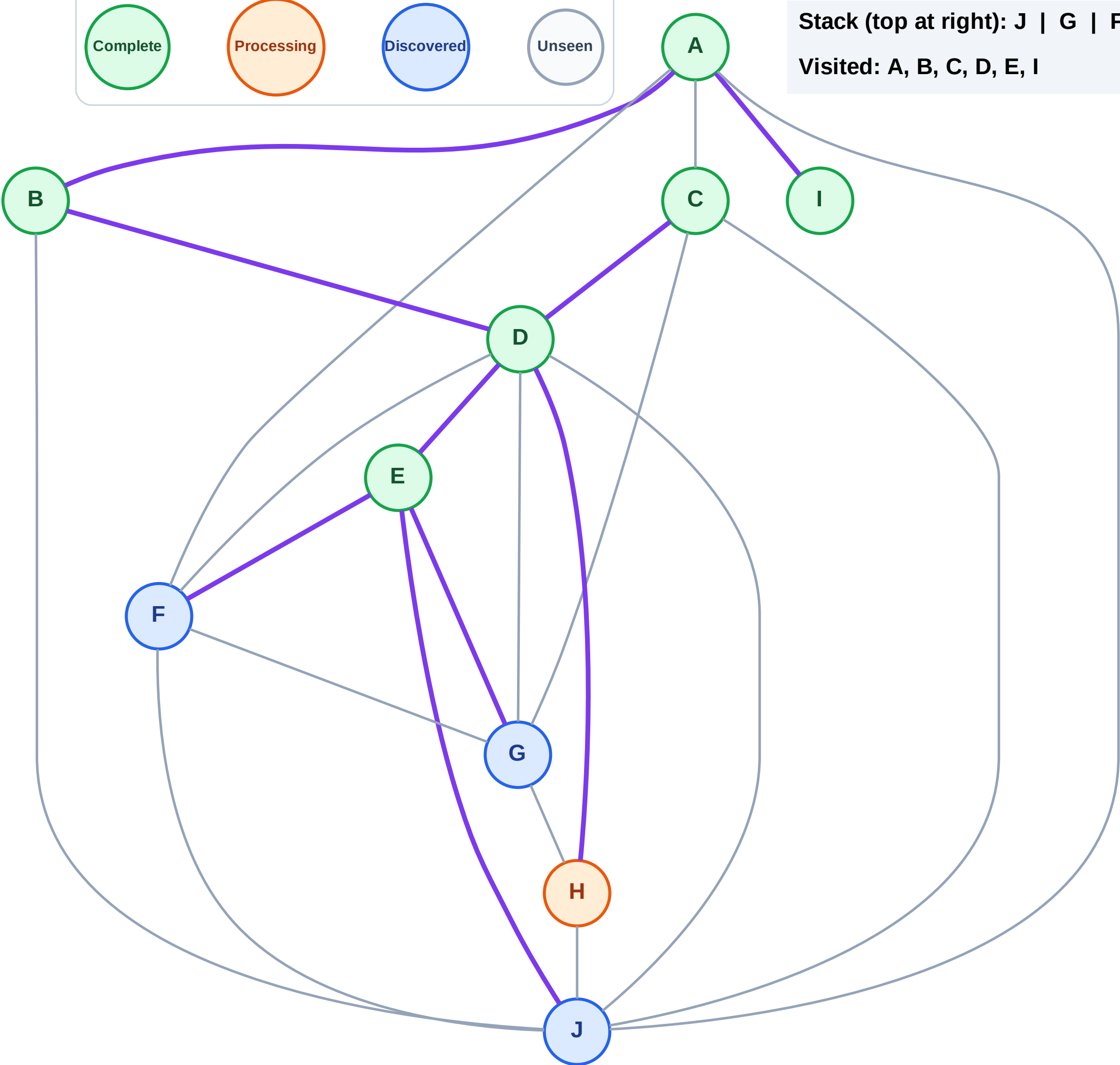
Complete

Processing

Discovered

Unseen

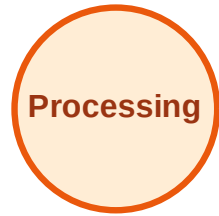
Stack (top at right): J | G | F
Visited: A, B, C, D, E, I



DFS Traversal

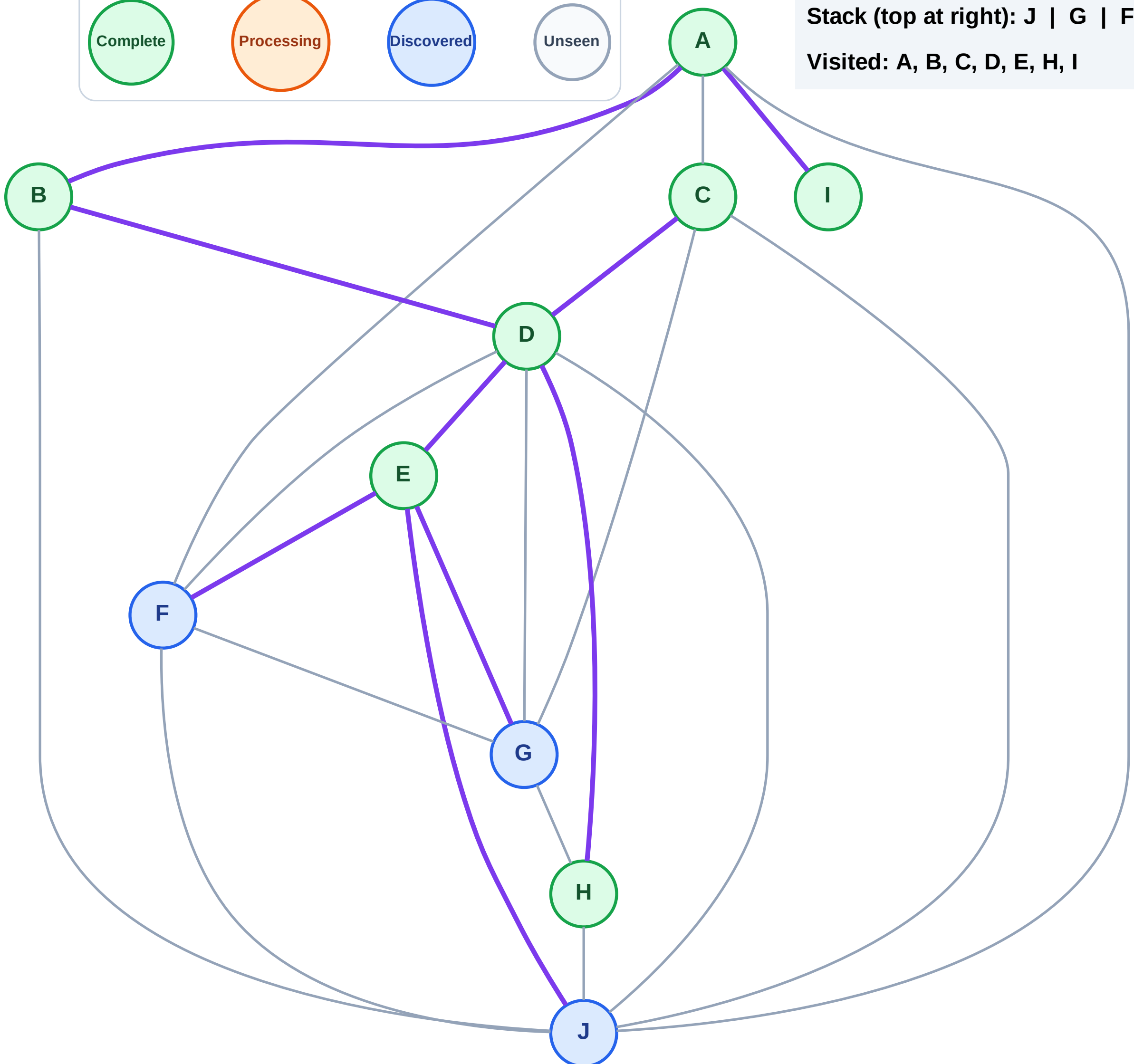
Step 15: No new neighbors from H

Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 16: Process node F

Legend

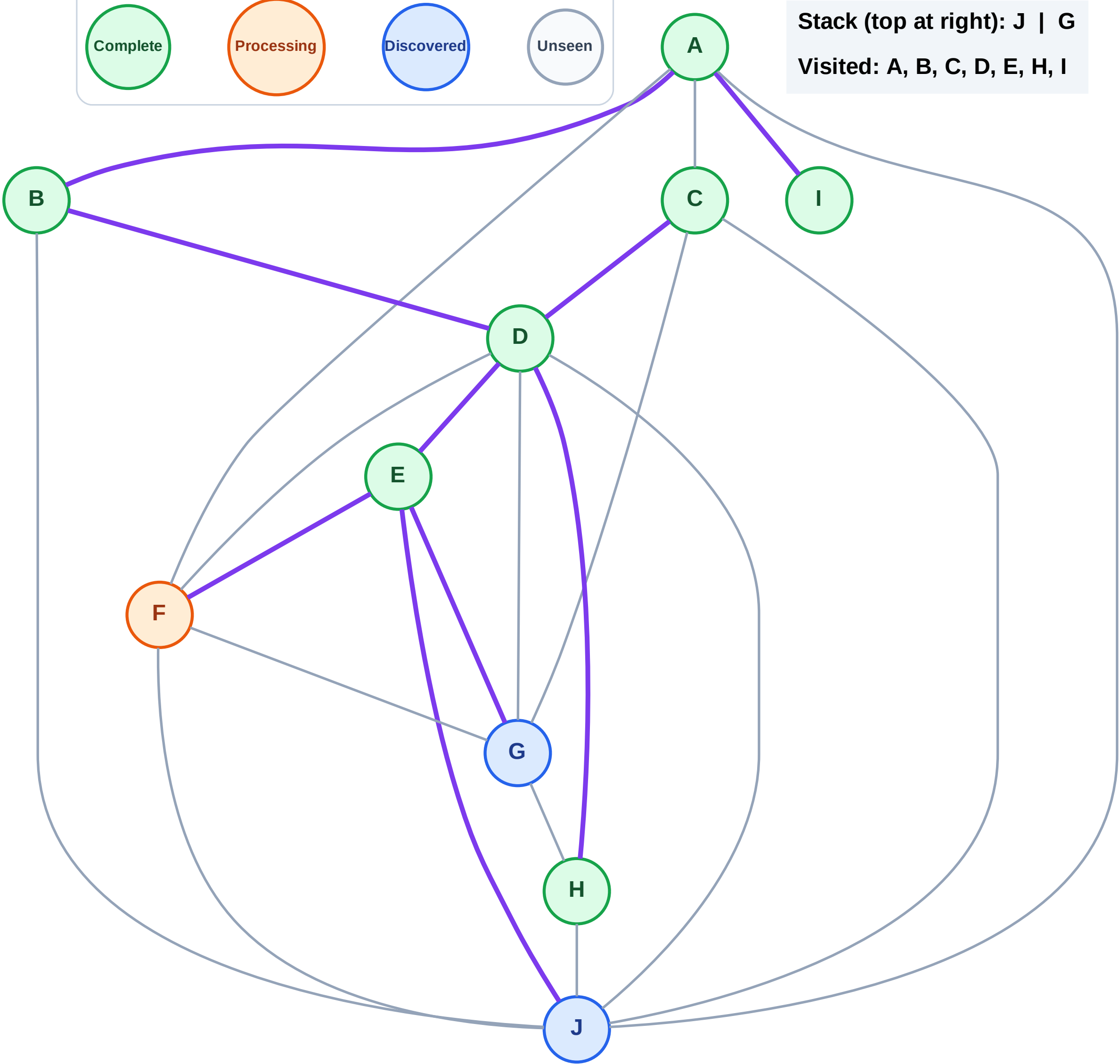
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G
Visited: A, B, C, D, E, H, I



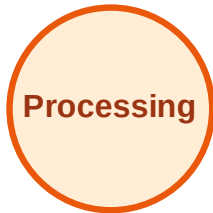
DFS Traversal

Step 17: No new neighbors from F

Legend



Complete



Processing



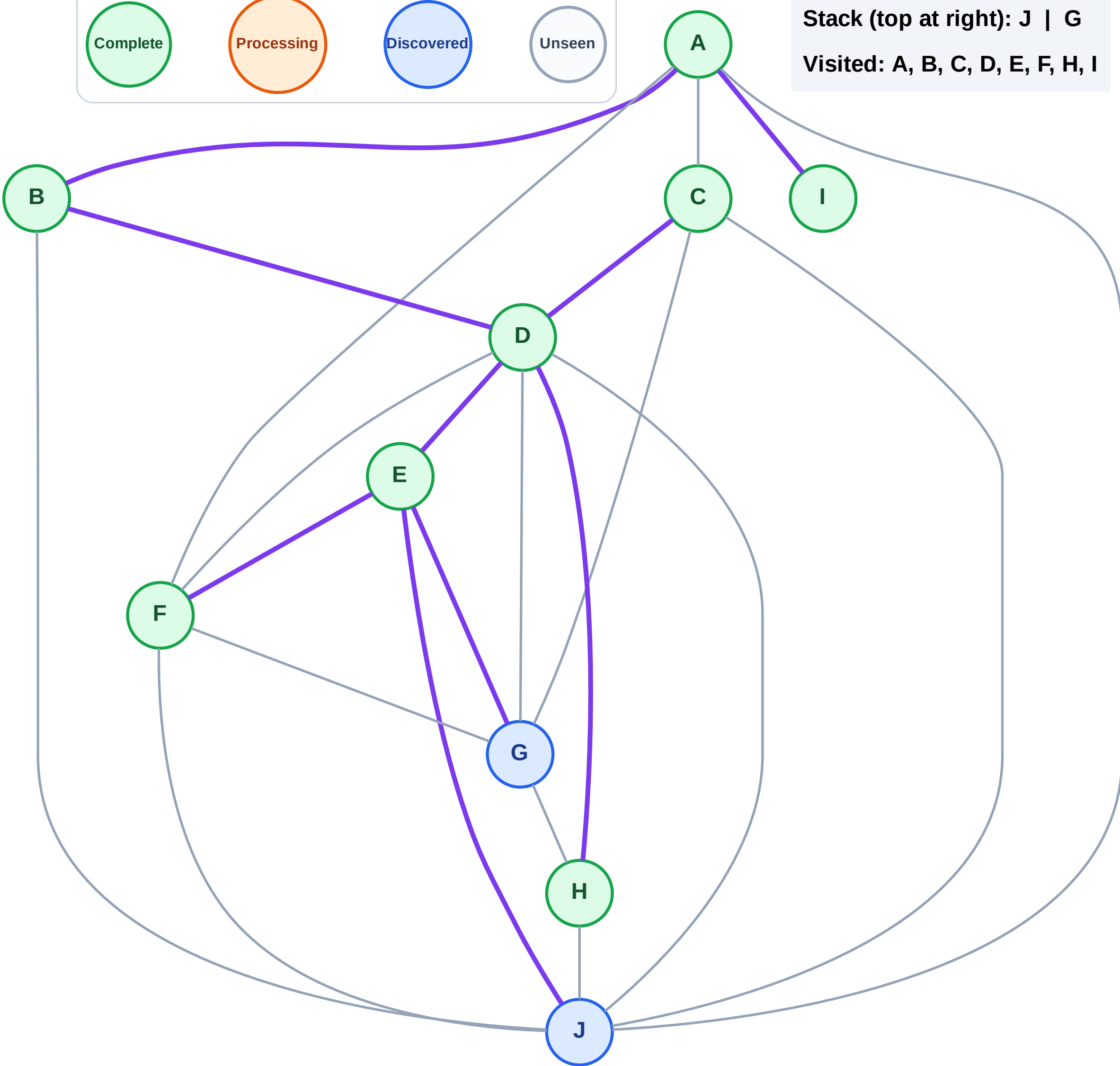
Discovered



Unseen

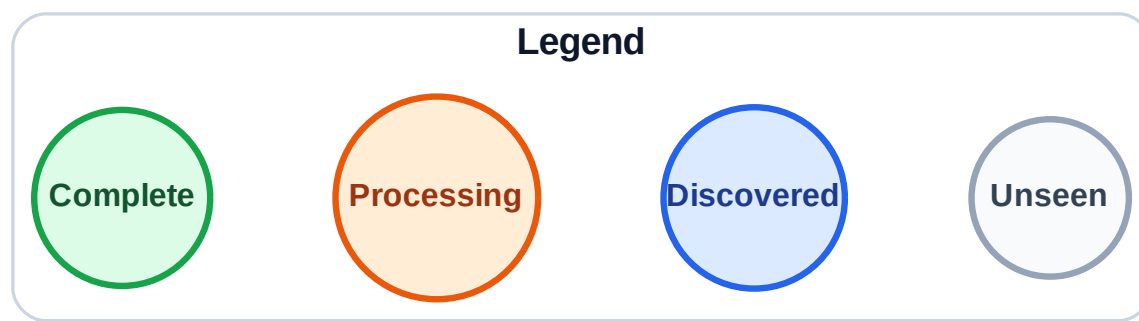
Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I

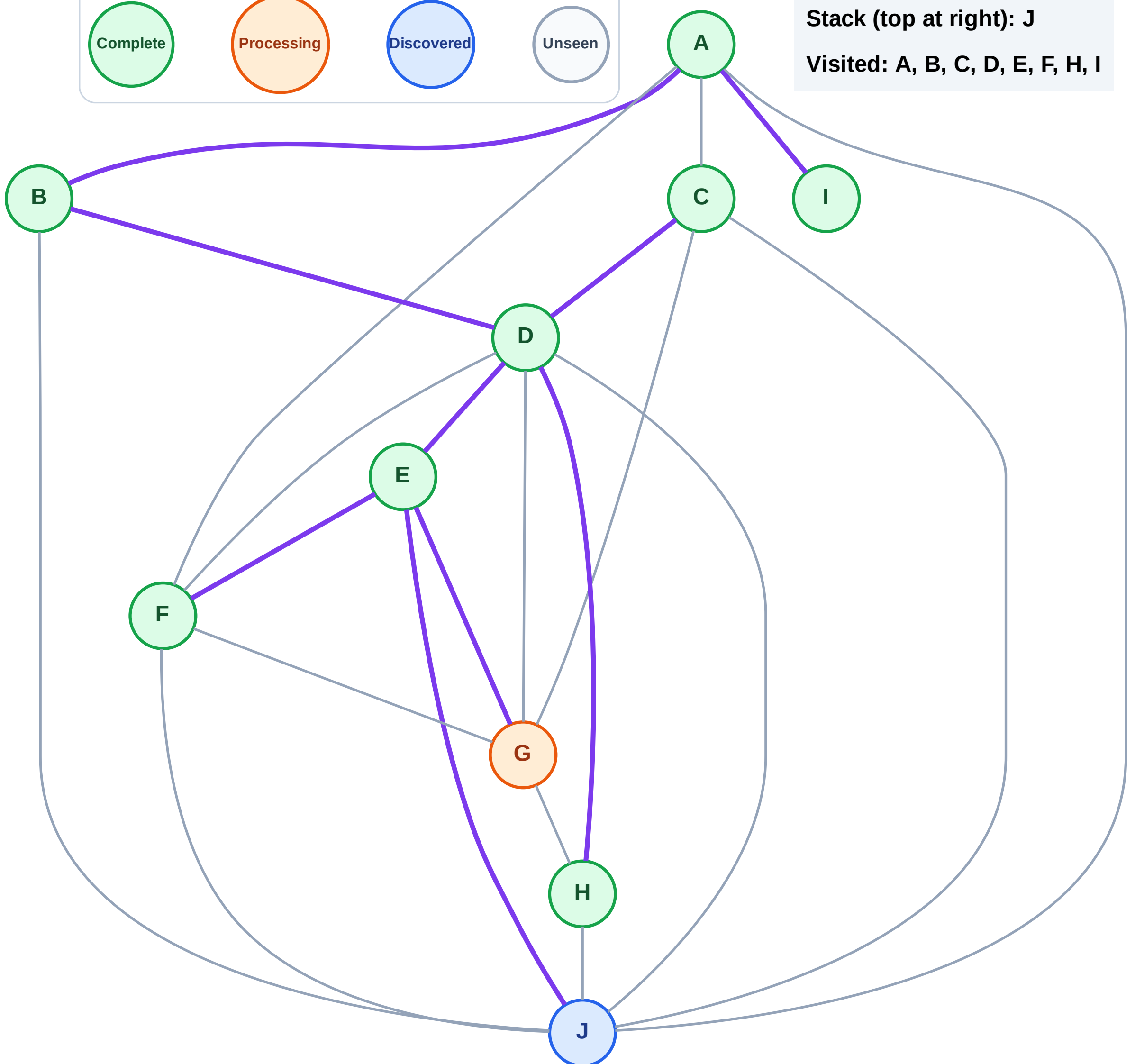


DFS Traversal

Step 18: Process node G



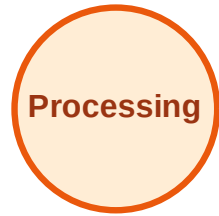
Stack (top at right): J
Visited: A, B, C, D, E, F, H, I



DFS Traversal

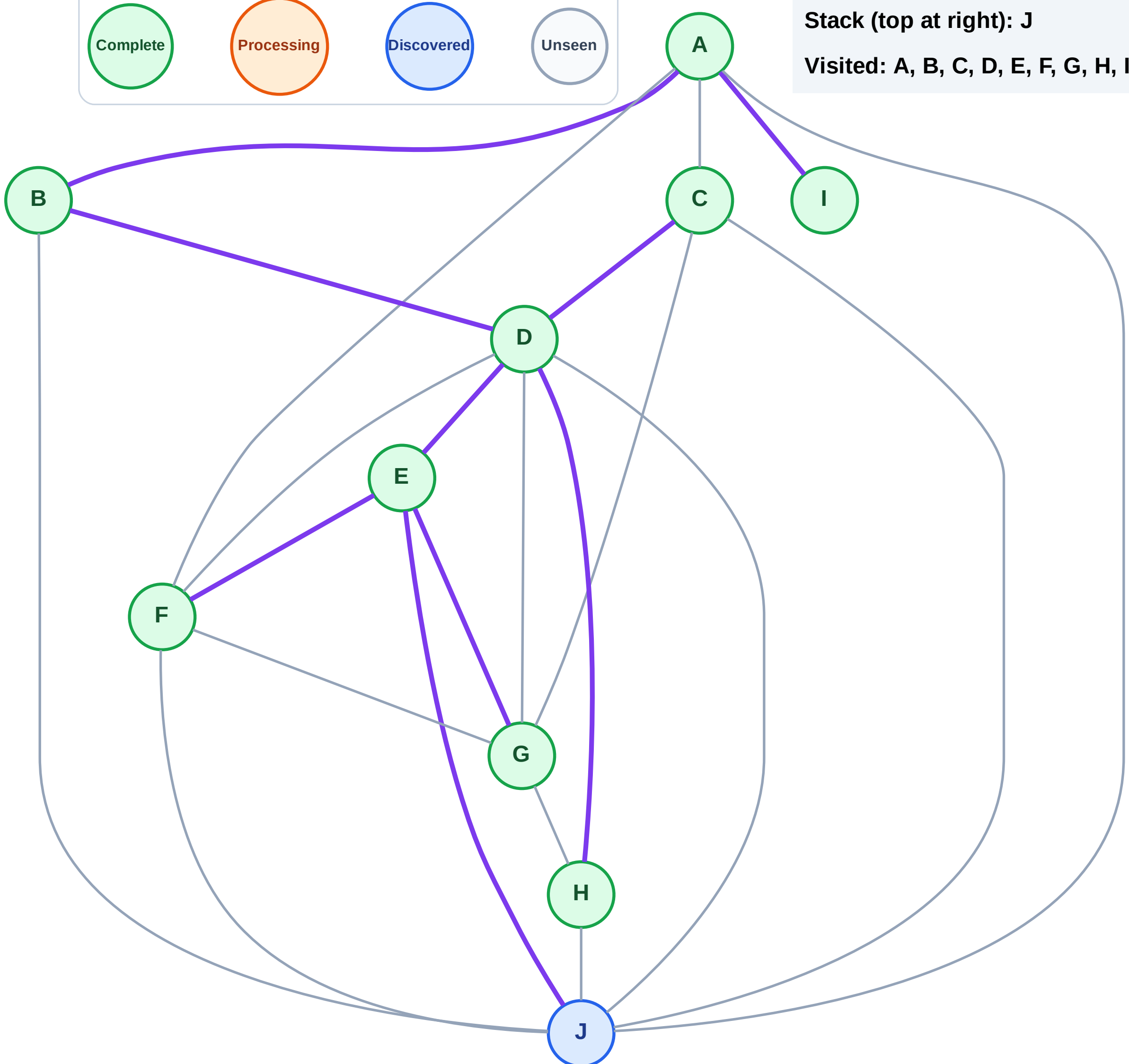
Step 19: No new neighbors from G

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

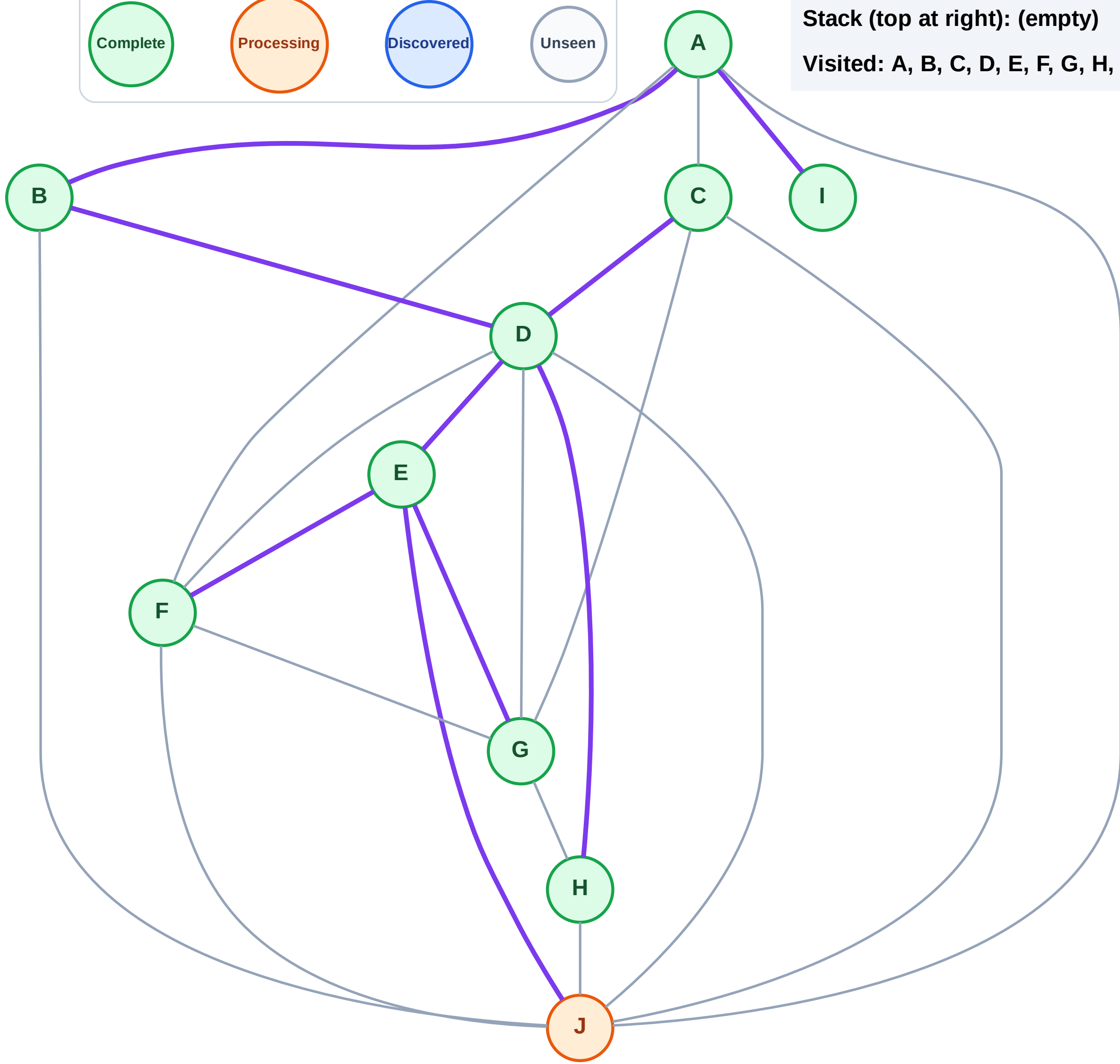
Complete

Processing

Discovered

Unseen

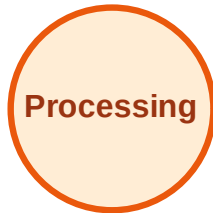
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

